Geometric Realization

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Abstract

We prove the Brown–Wemyss Numerical Realization Conjecture for completed Jacobi algebras of radical growth dimension at most one. The remaining potentials have two generators and cubic part x^3. For three families we construct a marked ADE arc and identify its contraction algebra. In the length-five family a filtered rank obstruction proves finiteness for every formal tail, and Wemyss's gluing theorem gives the contraction. Two growth estimates exclude the other branches. Lifted syzygies give a cubic lower bound when the reduced quartic vanishes, and partial morsification gives a cubic lower bound on each degenerate sextic axis. The only infinite-dimensional algebra in residual Type E satisfying the growth bound is $\mathrm{Jac}(x^3 + xy^3)$. We realize it by an explicit divisor-to-curve contraction.

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1 Introduction and the realization theorem

Which completed Jacobi algebras arise as contraction algebras of smooth threefolds? Brown and Wemyss conjectured that every completed Jacobi algebra with radical growth dimension at most one has a geometric realization. Both contraction types enter the question. A finite-dimensional algebra is to be realized by a flop, and an algebra of growth dimension one by a contraction of a divisor to a curve.

For a finite set of generators, put

$$F_d = \mathbb{C}\langle\langle x_1, \dots, x_d \rangle\rangle, \qquad \mathfrak{m}_d = (x_1, \dots, x_d), \qquad (F_d)_{\mathrm{cyc}} = F_d/\overline{[F_d, F_d]}.$$

The overline in the cyclic quotient denotes the closure of the *linear* commutator subspace. Throughout, every derivative ideal is understood to be a closed two-sided ideal. For a cyclic potential φ, let

$$A_\varphi = \mathrm{Jac}(\varphi) = F_d/\overline{(\partial_{x_1}\varphi, \dots, \partial_{x_d}\varphi)}, \qquad J_\varphi = \mathrm{rad}(A_\varphi).$$

For an augmented complete local algebra A, the radical growth dimension is defined by

$$\mathrm{Jdim}\, A = \inf\left\{r \geq 0 : \dim_{\mathbb{C}}(A/J^n) \leq C_r n^r \text{ for every } n \geq 1 \text{ and some } C_r > 0\right\}.$$

When the potential is fixed, we write A for its Jacobi algebra and J for its Jacobson radical. We write O_q for terms of ordinary word length at least q, indicating weighted orders explicitly when they are used. A necklace is a cyclic equivalence class of words. In each cyclic reduction, a necklace is good if it avoids the forbidden subwords, including across the cyclic boundary, and bad otherwise.

Definition 1.1. Write $\mathrm{Geo}(A)$ if there is a projective birational crepant morphism $\pi : X \to \mathrm{Spec}\, R$, where R is a three-dimensional complete local normal cDV complex algebra, X is regular, and the reduced closed fiber is $\mathbb{P}^1$, together with a continuous complete-algebra isomorphism $A \cong A_{\mathrm{con}}(\pi)$. Regularity is the smooth-source convention in this formal local setting. The generator used to define A_{con} is normalized and basic.

Conjecture 1.2 (Brown–Wemyss Numerical Realization)**.** *For every $d \geq 1$ and every cyclic formal potential $\varphi \in \mathfrak{m}_d^2$, the condition* $\mathrm{Jdim}\,\mathrm{Jac}(\varphi) \leq 1$ *implies* $\mathrm{Geo}(\mathrm{Jac}(\varphi))$.

A constant term in φ has zero cyclic derivatives. If its linear part is nonzero, one of its cyclic derivatives has nonzero constant term and is a unit. Thus we may work with potentials of order at least two.

Theorem 1.3 (Geometric Realization)**.** *Let φ be a cyclic formal potential in finitely many variables, with no linear term, and let its nonzero augmented Jacobi algebra satisfy* $\operatorname{Jdim}\operatorname{Jac}(\varphi) \leq 1$. *Then* $\operatorname{Jac}(\varphi)$ *is isomorphic to the normalized basic contraction algebra of a projective birational crepant contraction with regular source, complete local normal three-dimensional cDV target, and reduced closed fiber* $\mathbb{P}^1$. *The finite-dimensional case is flopping; the growth-one case is divisor-to-curve. In residual Type E the only growth-one algebra is* $\operatorname{Jac}(x^3 + xy^3)$.

The proof proceeds by the quartic jet. Section 2 constructs marked arcs of lengths three, four and six. When the quartic is alternating, the growth bound forces the coefficient of the pure fifth power to be nonzero. Section 3 then proves finiteness and applies the length-five construction. The branches left over have a vanishing reduced quartic or a degenerate pair of sextic coefficients. Section 4 excludes them by lower bounds that hold for every higher tail. Section 5 identifies the growth-one model and completes the proof.

Relation to earlier work

Donovan and Wemyss identify the contraction algebra of a flopping curve C with the algebra of noncommutative deformations of $\mathcal{O}_C(-1)$ [4, Theorem 1.2]. For the inverse problem, Brown and Wemyss proved realization in Types A and D and reduced the remaining case to two-variable potentials with cubic part x^3 [1, Conjecture 1.11, Corollaries 1.12 and 8.18].

Brown and Wemyss's classification of smooth threefold flops likewise begins with low-order normal forms [2, Section 3]. Their criterion uses *algebraic length*. One seeks a central element of the Jacobi algebra such that the quotient by the closed ideal it generates is isomorphic to $e\Pi e$, where Π is an ADE preprojective algebra and e is a primitive idempotent. The marked vertex corresponding to e determines the length [2, Definition 1.1, Corollary 8.14 and Theorem 9.2]. The numerical conjecture starts with the dimensions of the radical quotients alone.

For the family $x^3 + xy^3 + p(y)$, Pan constructs a smooth flop for every nonzero $p \in y^4\mathbb{C}[[y]]$ and describes the divisorial contraction at $p = 0$ [7, Proposition 4.11 and Corollary 4.12]. Brown and Wemyss also give a divisor-to-curve realization of $\operatorname{Jac}(x^3 + xy^3)$ [2, Theorem 1.7]. For lengths three, four and six our cyclic reductions use the higher coefficients of the given potential to construct an arc in the marked deformation space of Katz and Morrison [6, Sections 5–6]. Karmazyn's presentations of the universal flopping algebras [5, Theorem 1.3 and Examples 4.24–4.27] give the full algebra on this arc. For Van den Bergh's tilting generator [8, Propositions 3.2.6–3.2.7], one obtains the contraction quotient by killing maps that factor through the structure sheaf summand.

At length five we first prove that the Jacobi algebra is finite-dimensional, so we may pass from the formal potential to a polynomial with the same Jacobi ideal. The geometric input is the fifth family in Wemyss's gluing theorem [9, Section 1.6 and Theorem 1.7]. Brown and Wemyss describe the noncommutative deformation algebra of such curve neighborhoods by free necklace potentials [3, Corollary 6.4]. Proposition 3.9 also explains how the length-five normal form in [2, Theorem 8.3] verifies the geometric step. The filtered rank and gluing argument already gives this realization without invoking that normal form. For a vanishing reduced quartic and for either degenerate sextic axis we bound $\dim_{\mathbb{C}} A/J^N$ directly. The cubic lower bounds exclude these cases for every formal tail.

2 Reduction and constructive inputs

2.1 The published reduction

Proposition 2.1 (Published reduction)**.** *Conjecture 1.2 is equivalent to its two-generator restriction*

$$\varphi = x^3 + O_4, \qquad \operatorname{Jdim} \operatorname{Jac}(\varphi) \leq 1.$$

Cases of corank at most one and two-generator cases with $\dim J^2/J^3 = 2$ *are already geometric in both growth dimensions.*

Proof. The splitting lemma removes the quadratic variables without changing the completed Jacobi algebra [1, Lemma 4.5 and Appendix A]. Under $\operatorname{Jdim} \operatorname{Jac}(\varphi) \leq 1$, the corank bound leaves $\dim J/J^2$ at most two. In corank zero the algebra is $\mathbb{C}$, and in corank one it is $\mathbb{C}[[y]]$ or $\mathbb{C}[[y]]/(y^k)$.

In corank two the cubic part is nonzero. Up to a linear change of variables it is $x^3 + y^3$, x^2y, or x^3. The first two cubics give a second radical layer of dimension two, the third a layer of dimension three. The Type A and D constructions in [1, Sections 5–6 and Corollary 8.18] realize all the cases covered there in the sense of Definition 1.1, including the infinite-dimensional forms. Realization is invariant under continuous algebra isomorphism. It remains to treat the third cubic with an arbitrary tail, and the converse follows by specialization. □

2.2 Cyclic elimination and the quartic division

Lemma 2.2 (Removal of cyclic xx words)**.** *Every $x^3 + O_4$ is cyclically right-equivalent, by a continuous unitriangular automorphism fixing y, to a series whose higher necklaces contain no adjacent xx, including at the cyclic boundary. Its quartic part is*

$$a(xy)^2 + bxy^3 + cy^4. \tag{2.1}$$

Proof. At degree $n \geq 4$, rotate each necklace containing xx to the form x^2h, with h of degree $n-2$. Write the sum of these terms in degree n as $[x^2h_n]$. The substitution $x \mapsto x - h_n/3$, fixing y, changes the cyclic cubic $[x^3]$ by $-[x^2h_n]$ in this degree. Multiple replacements in the cubic and replacements in terms of degree at least four have degree greater than n. Each step therefore preserves all previous cancellations.

The substitutions and their inverses stabilize on every finite word quotient $F_2/\mathfrak{m}_2^k$. The inverse limit is the required automorphism. The three quartic necklaces in (2.1) exhaust those with no adjacent xx in cyclic order. The cyclic chain rule puts each derivative of the transformed potential in the image of the closed derivative ideal. Applying the inverse substitution gives equality of the ideals under transport. □

Proposition 2.3 (Five quartic orbits)**.** *Put $\Delta = b^2 - 8ac$. A linear map preserving the cubic has the form $x \mapsto \zeta x$, $y \mapsto tx + vy$, where $\zeta^3 = 1$ and $v \neq 0$. Its reduced quartic coefficients are*

$$\begin{aligned} a' &= v^2(\zeta^2 a + \zeta t b + 2t^2 c), \qquad & b' &= v^3(\zeta b + 4tc), \qquad & c' &= v^4 c, \\ \Delta' &= \zeta^2 v^6 \Delta. \end{aligned} \tag{2.2}$$

Allowing also a nonzero scalar multiple of the potential, the five four-jet representatives are

$$x^3, \quad x^3 + (xy)^2, \quad x^3 + xy^3, \quad x^3 + y^4, \quad x^3 + (xy)^2 + y^4.$$

Their strata are respectively the zero quartic, $c = b = 0, a \neq 0$, $c = 0, b \neq 0$, $c \neq 0, \Delta = 0$, *and* $c \neq 0, \Delta \neq 0$.

Proof. Expand the quartic and remove the cyclic xx terms by Lemma 2.2. Of the terms in $\zeta x(tx + vy)^3$ linear in t, just the middle placement of the new x contributes to $(xy)^2$, with coefficient $v^2\zeta tb$. The expansion of y^4 contributes the two words $xyxy$ and $yxyx$ to the same cyclic class, with total coefficient $2v^2t^2c$. This gives (2.2) and the stated transformation of the discriminant.

For $c = 0, b \neq 0$, take $t = -\zeta a/b$ to kill a'. For $c \neq 0$, take $t = -\zeta b/(4c)$ to kill b'. The remaining alternating coefficient is then proportional to $a - b^2/(8c) = -\Delta/(8c)$. Rescaling the variables and multiplying the potential by a nonzero scalar give the five representatives. Scalar multiplication preserves the derivative ideal. □

Remark 2.4. Each representative specifies only a four-jet. The higher terms in $\mathfrak{m}_2^5$ are arbitrary.

2.3 Geometry of marked arcs

For each length $\ell \in \{3, 4, 5, 6\}$, write $\mathbb{H}_\ell$ for Karmazyn's marked parameter ring, $\mathcal{Y}_\ell \to \operatorname{Spec} R_\ell$ for the simultaneous partial resolution, and Λ_ℓ for the full algebra [5, Theorem 1.3 and Section 3.3]. Let e_0 denote the projector onto the structure sheaf in Λ_ℓ. The universal contraction quotient is

$$C_\ell = \Lambda_\ell / \Lambda_\ell e_0 \Lambda_\ell.$$

Take an arc $\nu : \mathbb{H}_\ell \to \mathbb{C}[[w]]$ through the marked origin. Pull the family back along the arc, then localize and complete its affine target at the central singular point and base change the projective source to that target.

Lemma 2.5 (Endomorphisms and contraction quotients)**.** *For these marked families, the full endomorphism algebra commutes with arc base change and target completion. If the resulting source is regular and the morphism is crepant with reduced closed fiber* $\mathbb{P}^1$*, its normalized basic contraction algebra is isomorphic to the completed* e_0*-quotient.*

Proof. We use $\Lambda_\ell = \operatorname{End}(E)^{\mathrm{op}} = \operatorname{End}(E^\vee)$ for Karmazyn's specified tilting bundle $E = \mathcal{O} \oplus N$ [5, Theorem 3.3 and Section 3.3]. Both the universal source and the full algebra are flat over $\mathbb{H}_\ell$. On a finite affine cover of the separated source, the Čech complex for $\mathcal{E}nd(E)$ is bounded with flat terms. Its only cohomology is the flat endomorphism algebra in degree zero. Descending from the last term shows that every boundary module is flat. Tensoring the complex with the arc ring therefore computes the ordinary endomorphisms after pullback and gives zero higher self-Ext. Flat base change on the target gives the same comparison after localization and completion. The pulled-back bundle still generates, by adjunction and conservativity of pushforward for the affine base-change map to the original source.

On a crepant arc with regular source, rationality of the cDV target gives $R\pi_*\mathcal{O}_X = \mathcal{O}_{\operatorname{Spec} R}$, and the fibers have dimension at most one. The dual bundle $P = E^\vee$ retains its global generation, its relatively ample determinant and its structure sheaf summand. We have $H^1(P^\vee) = H^1(E) = 0$ because $E = \mathcal{H}om(\mathcal{O}, E)$ is a summand of $\mathcal{E}nd(E)$. The criterion of [8, Propositions 3.2.6–3.2.7] now identifies P as a projective generator for perversity minus one and E as a generator for perversity zero.

It remains to check the multiplicities of the summands. Modulo w they are the two distinct indecomposable summands of the basic generator on the completed central partial resolution.

Since w belongs to the Jacobson radical of the full algebra over the complete local target, a nonzero idempotent cannot reduce to zero. A decomposition upstairs would give a nontrivial decomposition on the central surface. An isomorphism between the projectives would also descend to that surface. Thus both summands occur once, and the structure sheaf still specifies the free projector.

The image of the natural multiplication map $\Lambda e_0 \otimes_H e_0\Lambda \to \Lambda$ is the two-sided ideal $\Lambda e_0 \Lambda$. Forming its cokernel commutes with arbitrary scalar extension, yielding

$$(\Lambda/\Lambda e_0\Lambda) \otimes_H S \cong (\Lambda \otimes_H S)/(\Lambda \otimes_H S)e_0(\Lambda \otimes_H S).$$

Let $\mathfrak{n}$ be the maximal ideal of the completed target and J the radical of the full algebra. Then $\mathfrak{n}\Lambda \subseteq J$. In the finite-dimensional algebra modulo $\mathfrak{n}$, the radical is nilpotent, giving $J^h \subseteq \mathfrak{n}\Lambda$ for some h. The two filtrations are therefore cofinal. The projector ideal is a finite submodule and is closed for this topology. Hence the geometric quotient agrees with the augmentation-complete quotient in the loop presentation. See also [Bi, Section 2.2] for the endomorphism comparison. □

Lemma 2.6 (The source first jet)**.** *For each residual A_{m-1} component, write its orthogonal projection polynomial as $p_m(Z, w) = Z^m + \sum_{i=0}^{m-2} a_i(w)Z^i$. The arc source is regular along the central fiber if and only if $a_0'(0) \neq 0$ at every such component.*

Proof. At each residual surface singularity the orthogonal projection of Katz and Morrison gives the local equation $UV = p_m(Z, w)$ after completion [6, Section 6, Theorem 3]. Its linear part is $-a_0'(0)w$. A nonzero derivative allows us to eliminate w, leaving three regular parameters. If the derivative vanishes, the three-dimensional local ring has embedding dimension four. The central surface is smooth elsewhere, so flatness over the arc gives regularity at its remaining points. The nonregular locus is closed and its image is closed by projectivity. A nonempty image would meet the closed point, which proves the criterion on the whole source. □

Proposition 2.7 (Geometry of the smooth arcs)**.** *The stated first-jet condition gives a projective birational crepant contraction with regular source, complete local normal cDV hypersurface target and reduced closed fiber $\mathbb{P}^1$. In the primitive length-four, length-five, and length-six families, the contraction is small. In length three, use the quotient parameters*

$$\beta^3 = B_1\beta + B_0, \quad \gamma^3 = C_1\gamma + C_0, \quad \delta^2 = D_0, \quad \beta + \gamma + \delta = t_K/3.$$

With B_0', C_0', D_0' nonzero at zero, it is divisorial precisely when

$$t_K = 0, \qquad B_1 = C_1, \qquad B_0 = -C_0. \tag{2.3}$$

Proof. The target is flat over $\mathbb{C}[[w]]$ and its central Cartier section is a Du Val hypersurface. Thus the target is a three-dimensional cDV hypersurface. To see that it is a domain, divide two nonzero elements by their respective maximal powers of w. Their reductions have nonzero product in the central fiber. At every smooth point of the central section the total space is regular. A codimension-one singular component would meet the section in positive dimension, although its singularity is isolated. This proves regularity in codimension one. The hypersurface satisfies S_2, so it is normal.

The partial resolution is projective and is an isomorphism over the smooth surface locus. Its scheme-theoretic closed fiber is preserved by base change and target completion. On the

central marked surface its reduction is $\mathbb{P}^1$ and its multiplicity is the coefficient ℓ of the retained vertex in the highest root [6, Sections 5–6]. Every component of the regular source meets this irreducible fiber by properness. Hence the source is integral, and the isomorphism over the smooth locus gives birationality.

Which roots can vanish along the arc? Work over $K = \mathbb{C}((w))$. For a deleted chain A_{m-1}, let $z_1, \dots, z_m$ be the centered roots of its projection polynomial. The first-jet hypothesis makes this polynomial Eisenstein. Set $w = q^m$, substitute $Z = qz$, and divide by q^m. The reduction is $z^m = c$ modulo q, with $c \neq 0$. Each simple root lifts uniquely, so the leading terms of the z_j in cyclic order are $c^{1/m}\zeta_m^j w^{1/m}$. Inertia permutes the m roots cyclically. In particular, every Levi root is a difference of distinct entries in one list and is nonzero along the arc.

Let $k > 0$ be the coefficient of the retained simple root in an ambient positive root. The value of this root along the arc decomposes as the sum of a base contribution in K and one contribution $\sum_j a_j z_j$ from each deleted Dynkin chain. The coefficients a_j are integers determined up to a common additive integer. If the chain attaches to the retained vertex at node h, these coefficients satisfy the congruence

$$\sum_j a_j \equiv -kh \pmod{m}.$$

On this chain the retained simple root pairs to -1 at the attachment vertex h and to zero elsewhere, so its restriction has weight $-\omega_h$. On a centered list the fundamental weight ω_h is represented by a sum of h entries. Adding a Levi root changes the coefficient vector by a difference of coordinate vectors. This gives the congruence, with the same divisibility tests in either orientation of the chain.

Suppose an arm degree m is prime and coprime to every other arm degree. If an ambient root vanishes, this arm's contribution lies in K, the intersection of its Puiseux field with the compositum of the other arm fields. Its Galois average is zero because the list is centered. Hence the contribution vanishes. The coefficient of $w^{1/m}$ in this contribution gives $\sum_j a_j \zeta_m^j = 0$. For prime m, the minimal polynomial of a primitive root of unity forces all the coefficients to be equal, so $m \mid kh$. For $m = 4$, a zero leading coefficient forces $a_1 = a_3$ and $a_2 = a_4$. Their sum must then be even. Applying these tests to the marked diagrams of [5, Examples 4.24–4.27] gives the table below, where $1 \leq k \leq \ell$.

ℓ	arm degrees	necessary conditions for persistence
3	$(2,3,3)$	$2 \mid k$, so $k = 2$
4	$(2,3,4)$	$3 \mid k$, so $k = 3$; the quartic leading term survives
5	$(4,5)$	$5 \mid k$, so $k = 5$; the quartic leading term survives
6	$(2,3,5)$	$2 \mid k$, $\quad 3 \mid k$, $\quad 5 \mid k$

The quartic chain in lengths four and five attaches at an end. An odd k gives an odd coefficient sum, so the term of order $w^{1/4}$ survives. In length four the other arms have no term of that order. In length five the quintic chain can be oriented with $h = 2$, coprime to five. Its contribution has already vanished, leaving the quartic term uncancelled. In length six no permissible coefficient meets all three divisibility tests. This excludes persistent roots in lengths four through six.

The exceptional case in length three. In the numbering of [5, Example 4.24], the retained vertex 6 is adjacent to $1, 3, 5$. To identify the remaining roots, start with $2\alpha_6 + \alpha_1 + \alpha_3 + \alpha_5$ in the

central D_4 subdiagram. Its projections to the two A_2 arms are extremal fundamental weights of squared norm $2/3$ when roots have squared norm two. Its projection to the A_1 arm is zero.

Every other coefficient-two root differs from this one by a Levi lattice vector, so its component perpendicular to the Levi span is fixed. In each relevant A_2 weight coset, the three permutations of the fundamental weight are exactly the minimum-norm centered triples. The representative attains the minimum in each arm. Any other root of the same total norm must do so too. Permute the triples independently to obtain all nine roots in a single Levi orbit.

Let b_i and c_j denote the roots of the two depressed cubics in the statement. In these centered coordinates, the vanishing conditions for the nine roots in the orbit become

$$-b_i - c_j = 2t_K/3.$$

Write t_i for the simple-root values in Karmazyn's coordinates. Substituting $t_K = t_0 + t_1/2$ and $3t_6 = -(t_0+2t_1+t_2+2t_3+t_4+2t_5)$ into the representative gives $-2t_K/3-(2t_2+t_3)/3-(2t_4+t_5)/3$. The last two fractions are entries of the centered cubic lists. Levi permutations give the other eight equations.

Averaging any one of these equations over cubic inertia gives $t_K = 0$. Since the cubics are irreducible, one opposite pair forces their root sets to be negatives of each other. Comparing coefficients gives (2.3). Conversely, those identities give three opposite pairs. Their three roots are orthogonal. For distinct pairs both entries change, and in each fundamental triple the inner product drops by one from the squared norm. Thus the pairing drops from two to zero, giving the generic configuration A_1^3.

If no root persists, the generic surface is smooth. The exceptional curves then lie in the central fiber, so the contraction is small. The target is normal and Gorenstein and there is no exceptional prime divisor, which gives crepancy.

In the exceptional length-three case the closures of the three generic root curves map onto curve germs. To see that these germs survive target completion, consider the relative critical algebra of the ADE surface equation. Its central fiber is the finite Jacobian algebra of the Du Val singularity. Homogeneous lifts of a basis generate the critical algebra over the polynomial marked base by induction on weight. The unique support point of the central fiber is the vertex. A split critical point therefore has integral coordinates on its trait and specializes to the vertex, so its map extends to the completion.

Every exceptional divisor dominates the parameter curve because the central source surface is integral and maps birationally to its image. On the generic surface, adjunction gives zero canonical intersection with each exceptional (-2)-curve. The intersection matrix is negative definite, so every discrepancy is zero over the splitting field and hence after descent. This proves crepancy. For the surface-deformation and root descriptions used above, see [6, Sections 5–6]. □

Remark 2.8. Flatness of the full algebra need not pass to its projector quotient. For example, the $\mathbb{C}[[w]]$-free order $\left(\begin{smallmatrix} \mathbb{C}[[w]] & \mathbb{C}[[w]] \\ w\mathbb{C}[[w]] & \mathbb{C}[[w]] \end{smallmatrix}\right)$ has free-vertex quotient $\mathbb{C}[[w]]/(w)$. Likewise, a nonzero finite local algebra on which a radical central element acts nilpotently cannot be flat over its parameter DVR. This is why we check endomorphism base change on the full algebra and then take the quotient.

2.4 Three-direction potentials and inverse preparation

Lemma 2.9 (Common central value)**.** *Let $P, Q, R \in u\mathbb{C}[[u]]$, and let U, V, W be their zero-constant antiderivatives. The completed Jacobi algebra of $F = U(x) + V(y) + W(-x - y)$ is the*

complete algebra with a central generator w and relations

$$P(x) = Q(y) = R(z) = w, \qquad x + y + z = 0.$$

Proof. The cyclic derivatives are $\partial_x F = P(x) - R(-x-y)$ and $\partial_y F = Q(y) - R(-x-y)$. In the Jacobi algebra, $P(x)$ and $Q(y)$ therefore have the common value $R(-x-y)$. It commutes with x as a series in x, and with y as a series in y. Since x and y topologically generate the algebra, this value is central. Adjoining $z = -x - y$ and $w = P(x)$ gives the displayed presentation. Eliminating the two new generators gives the inverse. Both maps are continuous for the augmentation-ideal topologies. □

Lemma 2.10 (One-variable preparation and its first jet)**.** *If $d \geq 2$ and $H(u) = h_d u^d + O(u^{d+1})$, $h_d \neq 0$, then $H(u) - w$ is a unit times a unique monic degree-d polynomial $M_H(u, w)$. Its lower coefficients lie in $w\mathbb{C}[[w]]$. After the shift $v = u + \rho_H(w)$, where $\rho_H = m_{d-1}/d$ and m_{d-1} is the coefficient of u^{d-1} in M_H, write*

$$M_H(v - \rho_H, w) = v^d - \sum_{i=0}^{d-2} T_{H,i}(w) v^i.$$

Then $T'_{H,0}(0) = 1/h_d$. If $\widetilde{H} - H \in u^M \mathbb{C}[[u]]$, $M \geq d+1$, then

$$\begin{aligned} \widetilde{\rho}_H - \rho_H &\in w^{\lfloor M/d \rfloor} \mathbb{C}[[w]], \\ \widetilde{T}_{H,i} - T_{H,i} &\in w^{\lceil (M-i)/d \rceil} \mathbb{C}[[w]]. \end{aligned} \tag{2.4}$$

Proof. Weierstrass preparation in $\mathbb{C}[[w]][[u]]$ writes $H(u) - w$ as a unit times a distinguished polynomial $M_H(u, w) \in \mathbb{C}[[w]][u]$ of degree d. To track its coefficients, set $w = q^d$. The d branches of $H(u) = w$ are $u_j = \zeta_d^j h_d^{-1/d} q + O(q^2)$, for $j = 0, \ldots, d-1$ and a primitive root ζ_d. The product $\prod_{j=0}^{d-1}(u - u_j)$ is fixed by $q \mapsto \zeta_d q$, so it belongs to $\mathbb{C}[[w]][u]$ and equals $M_H(u, w)$. Its constant coefficient is $-w/h_d + O(w^2)$. The shift by $\rho_H = m_{d-1}/d$ changes this coefficient by $O(w^2)$, giving $T'_{H,0}(0) = 1/h_d$.

Now perturb the series by an element of u^M. Substitute $u = qv$ and divide the equation by q^d. Its roots in v are simple modulo q, and the perturbation has order at least q^{M-d}. Each u_j thus changes by $O(q^{M-d+1})$. In the coefficient of u^i, the other $d - i - 1$ root factors contribute at least as many powers of q. The difference is divisible by q^{M-i}. Passing back to $w = q^d$ gives the ceiling in (2.4). For $i = d-1$ it gives the floor bound for ρ_H. The shift preserves the other bounds because ρ_H and every lower coefficient are divisible by w. Thus a potential correction of order $dN + 1$ changes the prepared coefficients only in $w^N \mathbb{C}[[w]]$, for any choice of complex coefficients. □

Theorem 2.11 (Realization of the three-direction families)**.** *Suppose the exact orders of P, Q, R are one of*

$$(3,3,2), \qquad (2,4,3), \qquad (2,3,5),$$

with nonzero leading coefficients. The Jacobi algebra of $U(x) + V(y) + W(-x-y)$ is geometric. The marked lengths are respectively three, four and six. The latter two are small and finite-dimensional. In length three the divisorial alternative is given by (2.3) on the prepared arc.

Proof. By Lemma 2.9 the three equations are $P(x) - w = 0$, $Q(y) - w = 0$ and $R(z) - w = 0$. Apply Lemma 2.10 to each equation and shift its generator to remove the next-to-leading coefficient. Their sum becomes $\beta + \gamma + \delta = \rho_P + \rho_Q + \rho_R$, so take $t_K = \ell(\rho_P + \rho_Q + \rho_R)$. The coefficients now give Karmazyn's quotient parameters [5, Section 3.3]. Removing the unit factors preserves the closed ideal, and the inverse shifts recover the original generators.

The leading coefficients of P, Q, R are nonzero. Lemma 2.10 therefore gives a nonzero $T'_{H,0}(0)$ on each arm. Proposition 2.7 constructs the contraction, and Lemma 2.5 identifies its normalized basic algebra. For a small contraction the quotient is finite over the target and supported at the closed point, so its complex dimension is finite. In length three apply (2.3) to the prepared coefficients. □

2.5 The nonzero quartic-discriminant stratum

Lemma 2.12 (Two cyclic tangent calculations)**.** *Give x, y weights $3, 2$, respectively, and set $r_x = 3x^2 + y^3$, $S = xy^2 + yxy + y^2x$. Every cyclic homogeneous polynomial of weight $d > 9$ has the form*

$$[ur_x + vS] + \lambda[y^{d/2}], \qquad \mathrm{wt}(u) = d - 6, \quad \mathrm{wt}(v) = d - 7,$$

where the last summand occurs only for even d. For $\lambda_0 \neq 0$, the ordinary graded quotient

$$D_{\lambda_0} = \mathbb{C}\langle x, y\rangle/(x^2, S + \lambda_0 y^3, [x, y^3])$$

has cyclic quotient spanned by $1, x, xy$ and pure powers of y. In degrees at least seven the pure classes vanish unless the degree is divisible by three. If $L_b = y + bx$ and $t_b = 1 - \lambda_0 b$, then

$$[L_b^{3k}] = t_b^k[y^{3k}], \qquad [L_b^5] = \frac{1 + 5t_b^2}{6}[y^5]. \tag{2.5}$$

Proof. Put $T = x^2 = -y^3/3$ in the quotient by r_x, S. It commutes with both generators. Over $\mathbb{C}[T]$ use the rules

$$xx \mapsto T, \qquad yyy \mapsto -3T, \qquad yyx \mapsto -xyy - yxy, \qquad yxyx \mapsto xyxy.$$

The last rule follows by comparing xS with Sx. Each rule decreases length or, at fixed length, lexicographic order with $x < y$. Their left sides have ten proper overlaps and no inclusions. The two reductions at each overlap agree, as the exact check over $\mathbb{Z}[T]$ in Section 6 verifies. The irreducible words are

$$1, y, y^2, x, xy, xy^2, xyx, xyxy, xyxy^2, yx, yxy, yxy^2.$$

These words form a basis over $\mathbb{C}[T]$. In the cyclic quotient, $[Sy^j] = 0$ gives $3[xy^{j+2}] = 0$ for $j \geq 0$, hence $T[x] = T[xy] = 0$. The cyclic class of xyx is $T[y]$. From $[Sx] = [Sxy] = 0$ we get $[xyxy] = -2T[y^2]$ and $2[xyxy^2] = -T[y^3]$. This accounts for the basis words and their central multiples. Above weight nine only pure powers of y can remain. Rotating each two-sided relation multiple to put the relation last gives the coefficient weights in the statement.

In D_{λ_0} put $Z = y^3$ and $q = xyx$. We have $[x, y^3] = 0$, so Z is central. Commuting x with $S + \lambda_0 Z = 0$ gives $x(S + \lambda_0 Z) - (S + \lambda_0 Z)x = 0$. Modulo $x^2 = 0$ this reads $xyxy = yxyx$,

which makes q central with $q^2 = 0$. The same twelve words span D_{λ_0} over $\mathbb{C}[Z]$. Multiplying $S + \lambda_0 Z$ successively by $1, y, y^2, x, yx$ and taking cyclic classes gives

$$[xy^2] = -\lambda_0[Z]/3, \qquad Z[x] = -\lambda_0 Z[y]/3, \qquad Z[xy] = -\lambda_0 Z[y^2]/3,$$
$$q[y] = (\lambda_0^2/3)Z[y], \qquad q[y^2] = (\lambda_0^2/6)Z[y^2].$$

The displayed identities reduce all twelve classes to pure powers of y and $1, x, xy$. Multiply the last two identities by q. Since $q^2 = 0$, it follows that $Z^2[y]$ and $Z^2[y^2]$ vanish. This gives the asserted vanishing in degrees at least seven.

For the powers of L_b, expand $L_b^3 = t_b Z + b^2 q$. Taking powers and using $[q] = 0$ gives the first identity in (2.5). Multiplying L_b^3 by L_b^2 and substituting the displayed trace identities gives the formula in degree five. In ordinary degree $n \geq 5$ any discrepancy therefore has the form $[\mu y^n + ux^2 + v(S + \lambda_0 y^3) + k[x, y^3]]$, where u, v, k have degrees $n-2, n-3, n-4$, respectively. □

Theorem 2.13 (Complete tails for $\Delta \neq 0$)**.** *Let $f = x^3 + O_4$ have reduced quartic $a_4(xy)^2 + b_4 xy^3 + c_4 y^4$. If $b_4^2 - 8a_4c_4 \neq 0$, then* $\mathrm{Geo}(\mathrm{Jac}(f))$. *The construction is a marked length-three E_6 arc. If $c_4 \neq 0$, it is small. If $c_4 = 0$, there is a cyclic right-equivalence to*

$$x^3 + xy^3 + p(y), \qquad p \in y^5\mathbb{C}[[y]],$$

and the construction is divisorial precisely for $p = 0$. All other members are finite-dimensional. The sole infinite-dimensional Jacobi algebra in this stratum, up to complete-algebra isomorphism, is $\mathrm{Jac}(x^3 + xy^3)$*, of growth dimension one.*

Proof. **The branch $c_4 = 0$.** When $c_4 = 0$, a shear and rescaling put the reduced quartic into the form xy^3. Eliminate the cyclic xx terms by Lemma 2.2. For weights $\mathrm{wt}(x, y) = (3, 2)$, every term beyond the cubic and quartic now has weight greater than nine. Apply the first calculation of Lemma 2.12 in increasing weight, retaining the residual pure powers of y. In weight d the changes $x \mapsto x - u$ and $y \mapsto y - v$ increase the generator weights by $d - 9 > 0$. Two replacements have weight at least $9 + 2(d - 9) > d$, so each cancellation survives the later steps. The substitutions and their inverses converge to give the form $x^3 + xy^3 + p(y)$.

To construct the geometric realization, set $\beta = y + x/2$ and $\gamma = -y + x/2$, and consider the three-direction potential

$$G_h = x^3 + \beta^4/4 - \gamma^4/4 + h(\beta), \qquad h \in u^5\mathbb{C}[[u]].$$

At $h = 0$ the cyclic class is $x^3 + xy^3 + x^3y/4$, so its first error has weight eleven. At each even weight $2m$ we may choose the coefficient of u^m in h to match the remaining class $[y^m]$. At odd weights there are only tangent terms. The same weight induction thus constructs both h and a convergent automorphism carrying G_h to the prescribed potential. In the generators β, γ the cubic is $x^3 = (\beta + \gamma)^3 = -(-\beta - \gamma)^3$, so the three one-variable derivatives have orders $(3, 3, 2)$. Theorem 2.11 gives the contraction.

The branch $c_4 \neq 0$. When $c_4 \neq 0$, choose τ satisfying $a_4 + b_4\tau + 2c_4\tau^2 = 0$. The resulting shear $y \mapsto y + \tau x$ eliminates the alternating quartic term, and gives $(b_4 + 4c_4\tau)^2 = \Delta \neq 0$. After rescaling coordinates and removing newly introduced cyclic xx quartics, we obtain $f_c = x^3 + xy^3 + cy^4 + O_5$ with $c \neq 0$. We then choose a root r of the quadratic equation

$$4cr(1-r) + 2r - 1 = 0, \quad L_1 = y + rx, \quad L_2 = -y + (1-r)x, \quad a = c(1-r) + 1/4, \quad d = cr - 1/4.$$

The quadratic equation excludes the values $r = 0, 1$ and yields the coefficients $a = (1-r)/(4r)$ and $d = -r/(4(1-r))$. The algebraic identities $a + d = c$, $4(ar - d(1-r)) = 1$, and $ar^2 + d(1-r)^2 = 0$ ensure that the linear combination $aL_1^4 + dL_2^4$ matches the target reduced quartic $xy^3 + cy^4$. All other degree-four terms are eliminated by a quadratic coordinate correction to x.

Apply the second calculation of Lemma 2.12 with $\lambda_0 = 4c$ and $t = 1 - 4cr = r/(1-r) \neq 0$. Powers of L_1 supply the residual class in degree $3k$ with coefficient t^k. In degree five, L_1^5 and L_2^5 have coefficients $(1 + 5t^2)/6$ and $-(1 + 5t^{-2})/6$. They cannot both vanish, since that would require both $t^2 = -1/5$ and $t^2 = -5$. Thus $h_1(L_1)$ and $h_2(L_2)$, starting in degree five, can match every residual class.

It remains to remove the relation multiples. Gradient corrections remove $[ux^2 + v(S + 4cy^3)]$. For $[k[x, y^3]]$, take the exponential of the derivation $D(x) = [k, x], D(y) = 0$. The exponential fixes the cyclic cubic x^3. The derivation raises word degree by $n - 4$, so the second variation of the quartic has degree $2n - 4 > n$. This permits induction one degree at a time. The coordinate maps and their inverses converge for arbitrary complex coefficients, giving the three-direction potential

$$x^3 + aL_1^4 + dL_2^4 + h_1(L_1) + h_2(L_2).$$

Because $L_1 + L_2 = x$, this three-direction potential has derivative orders $(3, 3, 2)$. The derivatives of the prepared constant coefficients for the two quartic arms are $1/(4a)$ and $1/(4d)$. Their sum is $c/(4ad) \neq 0$. Consequently, the length-three divisorial condition $B_0 = -C_0$ fails at first order, and the resulting contraction is small.

Finite dimension and growth one. Put $A_p = \mathrm{Jac}(x^3 + xy^3 + p(y))$. The element $T = x^2 = -y^3/3$ is central. For $p = 0$, the twelve normal words of Lemma 2.12 are a free basis over $\mathbb{C}[[T]]$ after completion. Their weights are bounded and the weight of T is six, so completing the normal forms amounts to completing their coefficients. The radical modulo T is nilpotent. Together with $T \in J^2$, this makes the two filtrations cofinal with linear bounds on the exponents and gives growth dimension one.

For $p \neq 0$, the initial relations for the weights $(3, 2)$ are still $3x^2 + y^3, S$. The associated graded algebra is a quotient of the unperturbed graded algebra. Lifting the twelve spanning words and reducing successively in the complete algebra gives at most twelve module generators for A_p over $\mathbb{C}[[T]]$.

Suppose $p \neq 0$ and the generic algebra is nonzero. A simple module over an algebraic closure of $\mathbb{C}((T))$ has dimension $r \leq 3$, since its matrix image has dimension $r^2 \leq 12$. Set $t^3 = T$. The eigenvalues of the diagonalizable matrix Y are $\lambda_i = \zeta_i t$, where $\zeta_i^3 = -3$. On the λ_i-eigenspace, the diagonal block of the second derivative relation reads $3\lambda_i^2 X_{ii} = -p'(\lambda_i)I$. Thus $\operatorname{tr} X \in \mathbb{C}((t))$. By $X^2 = t^3 I$, the same trace is an integer multiple of $t^{3/2}$. Its valuation forces it to be zero, so the two eigenvalues of X have equal multiplicity and $r = 2$.

Now write $p(y) = h_n y^n + \cdots$ with $h_n \neq 0$ and $n \geq 5$. The two contributions to the trace have leading sum proportional to $2\zeta_i^{n-3}$ for a repeated eigenvalue of Y, or to $\zeta_i^{n-3}(1 + \omega^{n-3})$ for two distinct ones, where $\omega^3 = 1$. Both are nonzero, contradicting $\operatorname{tr} X = 0$. The generic algebra therefore vanishes. A common power of T kills the twelve generators, so A_p is finite-dimensional.

A divisorial contraction in this marked family would have a nonzero quotient at the generic point of its exceptional image curve. On the split generic surface a generator summand pushes forward to a nonfree maximal Cohen–Macaulay module for the contracted (-2)-curve, and its identity survives in the stable endomorphism algebra. This rules out the divisorial alternative

for a finite-length algebra. When $p = 0$, Section 5 gives an explicit divisorial arc with contraction algebra A_0. Its infinite dimension already excludes a small contraction. □

2.6 Finite transfers and the length-four and length-six strata

For lengths four and six we must vary a one-variable summand while keeping the lower jets of the potential fixed. An exact flow supplies the desired cyclic class and controls the orders of both the change in the summand and the compensating change of coordinates.

Lemma 2.14 (Positive-order family transfer)**.** *Let $G = U(x) + V(y) + W(L)$, $L = y - \lambda x$, $\lambda \neq 0$. Suppose U' has exact order $d \geq 2$ and $W'(u) = bu^e + O(u^{e+1})$, $e > d$, $b \neq 0$. For $k \geq 1, s \geq 0$ with $r = d(k-1) + s \geq 2$ and for every $\rho \in \mathbb{C}$, there exist a modified U_ρ and an identity-tangent automorphism Ψ_ρ fixing y, such that*

$$\begin{aligned} \operatorname{ord}(U_\rho - U) &\geq dk + s, \\ \operatorname{ord}(\Psi_\rho(x) - x),\ \operatorname{ord}(\Psi_\rho^{-1}(x) - x) &\geq r, \\ [\Psi_\rho(U_\rho(x) + V(y) + W(L))] &= [G] + \rho b^k [L^{ek} x^s] + O_{ek+s+1}. \end{aligned}$$

The y-version uses V' of order d, fixes x, and adds $\rho b^k [L^{ek} y^s]$ in degree $ek + s$.

Proof. Introduce an indeterminate τ, set $A_\tau = U_\tau'(x)/\lambda$ and $B = W'(L)$, and solve the system

$$\begin{aligned} \partial_\tau U_\tau(u) &= \rho u^s (U_\tau'(u)/\lambda)^k, \\ C_\tau = \sum_{i=0}^{k-1} B^{k-1-i} x^s A_\tau^i, &\qquad D_\tau(x) = -\rho C_\tau/\lambda, \quad D_\tau(y) = 0. \end{aligned}$$

Since $r = d(k-1) + s \geq 2$, each step of the coefficient recursion increases the index by at least $r - 1 > 0$. Thus the coefficients of U_τ are polynomials in τ. The order of C_τ is at least r, so the same recursion solves $\partial_\tau \Psi_\tau = \Psi_\tau D_\tau$ with $\Psi_0 = \mathrm{id}$ and gives coefficients of Ψ_τ polynomial in τ. On every $F_2/\mathfrak{m}_2^N$ the resulting substitution is unipotent. Its inverse is polynomial in the time parameter and satisfies the same order bound. We may therefore evaluate at $\tau = 1$ compatibly on all finite quotients.

Noncommutative telescoping and cyclic rotation in the cyclic quotient give

$$[x^s A_\tau^k] - [B^k x^s] = [C_\tau (A_\tau - B)], \quad \partial_\tau [\Psi_\tau(G_\tau)] = \rho [\Psi_\tau(B^k x^s)].$$

The degree-$ek+s$ term on the right-hand side equals $\rho b^k [L^{ek} x^s]$ throughout the flow. Integrating these polynomial coefficients from $\tau = 0$ to $\tau = 1$ establishes the assertion for the first assertion. For the y-branch, we set $A_\tau = -V_\tau'(y)$, $\partial_\tau V_\tau = \rho u^s (-V_\tau')^k$, $C_\tau = \sum_{i=0}^{k-1} B^{k-1-i} y^s A_\tau^i$, and $D_\tau(y) = \rho C_\tau$, leaving x fixed. The exact flow identity then yields the asserted leading variation and order estimates. □

Lemma 2.15 (The weighted y^4 reduction)**.** *Every $x^3 + y^4 + O_5$ is right-equivalent, with the entire tail retained, to*

$$x^3 + y^4 + \alpha_{17} xyxy^2 + \alpha_{20} (xy^2)^2 + \alpha_{21} (xy)^3 + O_{\mathrm{wt}}(24), \qquad \mathrm{wt}(x, y) = (4, 3),$$

where every remaining necklace avoids the cyclic subwords xx and yyy. The corresponding ordinary remainder has order at least seven.

Proof. In weighted degree $n > 12$, rotate each bad necklace to place a forbidden subword first. Their sum can then be written cyclically as x^2u_n plus y^3v_n. The substitutions $x \mapsto x - u_n/3$ and $y \mapsto y - v_n/4$ raise generator weights by $n - 12 > 0$. They cancel these terms to first order and affect the potential nonlinearly only above weight n. The successive maps and their inverses therefore converge.

The tail begins in ordinary degree five, so the steps at weights thirteen and fourteen are trivial. In particular they introduce no $(xy)^2$ term. Each surviving higher necklace is made up of blocks xy of weight seven and xy^2 of weight ten, with at least two occurrences of x. Listing the block words of weight below twenty-four gives the three displayed terms of weights seventeen, twenty and twenty-one. All later possibilities have ordinary length at least seven. □

Theorem 2.16 (Length-four realization)**.** *For every $a \neq 0$ and every cyclic formal tail $h \in O_6$,*

$$f = x^3 + y^4 + a\,xyxy^2 + h$$

is right-equivalent to $x^3 + V(y) + W(y - x)$, where $V(u) = u^4 + O(u^5)$, $W(u) = (a/5)u^5 + O(u^6)$. Its Jacobi algebra is finite-dimensional and geometric through a smooth marked length-four small contraction.

Proof. **Residual cyclic classes.** Write $u = xy$, $v = xy^2$, $g = xyxy^2$ and $q_{17} = [x, \partial_x g]$. To find the degree-$m$ errors that survive the principal corrections and conjugation, consider

$$D_{17} = \mathbb{C}\langle x, y\rangle/(x^2, y^3, q_{17}).$$

Every surviving mixed necklace is a cyclic word in the blocks u, v. We claim that for $m \geq 6$, it suffices to keep $[u^{m/2}]$ when m is even, together with $[u^{2k-2}v^2]$ when $m = 4k + 2$. All odd degrees in this range vanish. Modulo x^2, y^3, differentiation gives

$$q_{17} = uv + vu - yvx - y^2ux.$$

Every nonempty word H in u, v begins with x. Thus $[Hyvx]$ and $[Hy^2ux]$ vanish because xx occurs across the cyclic boundary. This gives $[Huv] = -[Hvu]$, so in a cyclic word with at least three blocks an interchange of unlike neighbors changes the sign. Modulo the monomial relations, $q_{17}y^2 = -yv^2 - y^2uv$. Multiply on the left first by y, then by x, to get $v^3 = 0$ in D_{17}. Sorting therefore kills every class with at least three occurrences of v.

For a sorted mixed word u^iv^j with at least three blocks, rotate one u and commute it back past the v's. The sign is $(-1)^j$. Moving one v in the same way gives $(-1)^i$, so both exponents must be even. The only mixed two-block word has degree five. This proves the spanning assertion for $m \geq 6$. Rotating each two-sided relation multiple gives the remaining terms as $[Hq_{17} + x^2u_m + y^3v_m]$, with coefficient degrees $m - 5, m - 2, m - 3$.

Matching degree six. In the seed $x^3 + y^4 + (a/5)(y - x)^5$ the unique good fifth necklace has coefficient a. Remove the bad fifth necklaces by principal x- and y-corrections. Write $L = y - x$. To adjust the sixth jet, apply the y-flow of Lemma 2.14 before those normalizing substitutions, with $(d, e, k, s) = (3, 4, 1, 2)$ and amplitude $-A/4$. The leading changes are Ay^5 in V and $-Ay^2/4$ in the coordinate. By the exact flow identity their net change in degree six is $-aA[L^4y^2]/4$. The normalizing substitutions preserve this term because they are tangent to the identity.

In L^4y^2 only xy^2xy^2 survives cyclically. For the other direction add $B\xi^6$ to the one-variable function $W(\xi)$. This produces $B[L^6]$, containing three rotations of v^2 and two of u^3 with opposite signs. Together the changes give

$$(-aA/4 + 3B)v^2 - 2Bu^3 \tag{2.6}$$

in the cyclic quotient. The determinant is $a/2$, so both sixth-degree coefficients can be matched.

Higher degrees and convergence. Assume agreement below degree $m \geq 7$. At $m = 4k$, changing $W(\xi) \mapsto W(\xi) + \mu\xi^m$ contributes $2\mu[u^{2k}]$. At $m = 4k + 2$ the same change has coefficient -2μ on $[u^{2k+1}]$. The second direction comes from the y-flow with exponent k and $s = 2$. Its leading term is $\rho a^k[L^{4k}y^2]$.

The appended y^2 forces the last block to be v after rotation. For a word with two v's, the other doubled gap may occupy any of the first $2k - 1$ positions. Moving the corresponding v's together gives alternating signs with sum one. Words with more than two v's vanish, and the appended pair excludes the pure u-class. Hence the flow contributes $\rho a^k[u^{2k-2}v^2]$, as required. In odd degrees there is no residual class to match. At every stage the flow acts on the current family, with order bounds $3k + 2$ and $3k - 1$.

To remove the term $[Hq_{17}]$, send x to $\exp(H/a)x\exp(-H/a)$, fixing y. Conjugation preserves the cubic cyclically and fixes the pure quartic. The first variation of ag is $[Hq_{17}]$, and the second has degree $2m - 5 > m$. Principal corrections in x, y remove the multiples of x^2, y^3, with the signs chosen to cancel the error. The changes of the family, the coordinate maps and their inverses stabilize on every finite word quotient. Passing to the limit and taking $X = x, Y = y - x$ gives derivative orders $(2, 4, 3)$. Theorem 2.11 yields the claimed smooth length-four contraction. □

Theorem 2.17 (Nonzero sextic-pair realization)**.** *For $b, c \neq 0$, every*

$$f = x^3 + y^4 + b(xy^2)^2 + c(xy)^3 + O_7$$

is right-equivalent to $x^3 + V(y) + W(y - \lambda x)$, where

$$\lambda = -3c/(2b), \quad d_0 = b/(3\lambda^2), \quad V(u) = u^4 + O(u^5), \quad W(u) = d_0u^6 + O(u^7).$$

It is realized by a smooth marked length-six small contraction and has finite-dimensional Jacobi algebra.

Proof. Set $L = y - \lambda x$, $g_6 = b(xy^2)^2 + c(xy)^3$, $q_6 = [x, \partial_x g_6]$, and $D_6 = \mathbb{C}\langle x, y\rangle/(x^2, y^3, q_6)$. To find the cyclic classes that must be matched in each degree, we first exhibit a central element of degree five. In the quotient by x^2, y^3, the operation $h \mapsto [x, \partial_x h]$ kills every bad necklace. For x^2a, differentiation of the pair gives $[x, xa + ax] = [x^2, a]$, and all other terms retain x^2. A representative y^3a retains y^3 under every x-derivative. The good coefficients of d_0L^6 are $3d_0\lambda^2 = b$ and $-2d_0\lambda^3 = c$. Consequently

$$q_6 = -6\lambda d_0[x, L^5] \pmod{(x^2, y^3)}.$$

Since $[y, L^5] = \lambda[x, L^5]$, the element $z = L^5$ is central in D_6.

After rescaling x by λ, the quotient by z is

$$B_6 = \mathbb{C}\langle x, y\rangle/(x^2, y^3, (y - x)^5).$$

To compute the graded algebra, index the columns in degree n by words avoiding xx, yyy. For each pair of such words a, b with $|a| + |b| = n - 5$, expand $a(y - x)^5 b$ and discard the forbidden words. The resulting rows span the degree-n relations. For the cocenter, identify cyclic rotations and discard any class with a forbidden rotation. Now the rows $[(y - x)^5 b]$ suffice, since a left multiplier rotates to the right. Gaussian elimination over the rationals gives, in degrees zero through fifteen,

$$\dim(B_6)_n : 1, 2, 3, 4, 5, 6, 6, 6, 6, 6, 5, 4, 3, 2, 1, 0,$$
$$\dim HH_0(B_6)_n : 1, 2, 2, 1, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0.$$

Adding the rows of $1, y, y^2, L, L^2, L^3, L^4$ in the indicated degrees gives full column rank. The script in Section 6 verifies the ranks. Degree fifteen vanishes in the algebra itself, so every longer word vanishes as well.

Centrality gives $HH_0(D_6)/zHH_0(D_6) = HH_0(B_6)$. Subtract a linear combination of the seven lifts from a homogeneous class. The remainder belongs to $zHH_0(D_6)$, hence is z times a class of degree five less. Induction now gives generation over $\mathbb{C}[z]$ in every degree.

Thus in degree $m \geq 7$ the residual classes are spanned by $[L^m]$ and, if $m = 5k + j$ with $j = 1, 2$, by $[L^{5k} y^j]$. After these classes have been matched, the degree-m discrepancy has the form $[x^2 u_m + y^3 v_m + H_m q_6]$, where the coefficient polynomials have degrees $m - 2, m - 3, m - 6$, respectively.

The seed $x^3 + y^4 + d_0 L^6$ has the required coefficients $3d_0\lambda^2$ and $-2d_0\lambda^3$. Remove its bad sixth-degree necklaces by principal corrections in x and y. Suppose the current family agrees with the prescribed potential through degree $m - 1$. Varying the coefficient of degree m in W supplies L^m. To supply $L^{5k} y^j$, use the second flow in Lemma 2.14 with $d = 3, e = 5, s = j$. Divide the required coefficient by $(6d_0)^k$ to obtain the amplitude ρ.

The order condition $r = 3(k - 1) + j$ is already at least two in the first relevant degrees. The least k is two for $j = 1$ (degree eleven) and one for $j = 2$ (degree seven). The exact identity of Lemma 2.14 thus applies from the first step, changing V only from order $3k + j$.

Finally, send x to $\exp(H_m) x \exp(-H_m)$, fixing y. The resulting first variation supplies $[H_m q_6]$ and the second variation on g_6 has degree $2m - 6 > m$. Principal corrections in x and y remove the other relation multiples. Each previously matched degree is preserved. As m increases, the orders of the family updates, maps and inverse maps tend to infinity, so their limits give the desired right-equivalence. With $X = \lambda x$ and $Y = -y$, the three derivative orders are $(2, 3, 5)$. Theorem 2.11 now gives a smooth length-six small contraction. □

3 Realization of the length-five stratum

For the alternating quartic, the bound $\mathrm{Jdim} \leq 1$ forces the pure fifth coefficient to be nonzero. We first normalize this coefficient, then prove that every algebra with the resulting five-jet is finite-dimensional. The geometric realization follows by truncation and gluing.

3.1 The alternating quartic and its fifth jet

Theorem 3.1 (A uniform weighted growth obstruction)**.** *If every word of f has weight at least six for $\mathrm{wt}(x, y) = (2, 1)$, then for every $N \geq 1$*

$$\dim_{\mathbb{C}}(\mathrm{Jac}(f)/J^N) \geq \#\{(a, b, c) \in \mathbb{Z}^3_{\geq 0} : 2a + b + 3c < N\} \geq \big(\lfloor (N - 1)/6 \rfloor + 1\big)^3. \tag{3.1}$$

In particular $\mathrm{Jdim}\,\mathrm{Jac}(f) \geq 3$.

Proof. Let K_N be the closed span of words of weight at least N. A cyclic derivative reduces weight by at most two, so only coefficients in weights six through $N+1$ enter the quotient. Vary them independently about $f_0 = (xy-yx)^2/2$, with perturbation $g = \sum e_i h_i$ over $T = \mathbb{C}[[e_1, \dots, e_r]]$. Adjoin ζ of weight three. The cyclic derivatives of $\zeta(xy-yx) - \zeta^2/2 + g$ give the rules

$$yx \longmapsto xy - \zeta, \quad \zeta y \longmapsto y\zeta + \partial_x g, \quad \zeta x \longmapsto x\zeta - \partial_y g.$$

Truncate first in parameter degree. Each derivative correction increases that degree. At fixed parameter degree the remaining terms decrease word length or, at fixed length, the number of inversions for $x < y < \zeta$. This proves termination, and none of the rules decreases weight. The only overlap is ζyx. Its two reductions differ by $-[y, \partial_y g] - [x, \partial_x g]$, which vanishes by the cyclic identity $[x, \partial_x g] + [y, \partial_y g] = 0$. Disjoint reductions commute.

The ordered monomials $x^a y^b \zeta^c$ with $2a + b + 3c < N$ therefore give unique normal forms at every parameter cutoff. Passing to the inverse limit gives a finite free T-module on those monomials. Eliminating $\zeta = xy - yx$ identifies it with the weighted quotient of the original Jacobi algebra.

The finite relation matrix has polynomial entries in e_i. Its generic rank is the dimension of the word space minus the number of ordered monomials. Every larger minor vanishes as a power series at the seed and hence as a polynomial. Specialization can only decrease the rank. Translating the coefficients by f_0 thus gives the same dimension lower bound for every potential of weight at least six. At a fixed cutoff the derivative ideal and its closure have the same image, so formal tails cause no further relations there.

Since $J^N \subseteq K_N \operatorname{Jac}(f)$, the radical quotient surjects onto the weighted quotient. This gives the first inequality of (3.1). For the second, bound each of the three nonnegative exponents by $\lfloor (N-1)/6 \rfloor$. Their weight then satisfies $2a + b + 3c \le 6\lfloor (N-1)/6 \rfloor < N$. □

Corollary 3.2 (The alternating-quartic five-jet)**.** *If $f = x^3/3 + (xy)^2/2 + O_5$ and $\operatorname{Jdim} \operatorname{Jac}(f) \le 1$, its pure y^5 coefficient is nonzero. A formal coordinate change and a nonzero scalar multiple put its five-jet into*

$$x^3/3 + (xy)^2/2 - xy^4 - y^5/5.$$

Proof. The only cyclic necklaces of degree five avoiding adjacent xx are $xyxy^2, xy^4$, and y^5. If the coefficient of y^5 were zero, every surviving word in the potential would have $(2,1)$-weight at least six, which by Theorem 3.1 would force $\operatorname{Jdim} \operatorname{Jac}(f) \ge 3$, contradicting $\operatorname{Jdim} \le 1$. Let the coefficients of $xyxy^2, xy^4, y^5$ be denoted by A, B, C, respectively, with $C \neq 0$. Rescaling $x \mapsto s^2 x$ and $y \mapsto sy$, and multiplying the overall potential by s^{-6} with $s = -5C$, gives $C = -1/5$. We then apply the shear $y \mapsto y + dx$ and restore the standard four-jet via the quadratic correction $x \mapsto x - dxy - d^2x^2/2$. The shear shifts the coefficient B of the one-x necklace xy^4 to $B - d$, while the restoring substitution alters only necklaces containing at least two x's. Setting $d = B + 1$ normalizes this coefficient to -1. We then eliminate the bad fifth necklaces using a cubic correction in x, and finally apply the quadratic shift $y \mapsto y - A'y^2$ to cancel the residual $A'xyxy^2$ term. The remaining terms have order at least six. □

3.2 The length-five finiteness theorem

For the finiteness calculation we use variables of weights one and two. Proposition 3.9 relates them to the preceding five-jet by an explicit change of coordinates. Let

$$p(x) = a_4 x^4 + O(x^5), \qquad a_4 \neq 0, \qquad I_p' = p, \quad I_p(0) = 0,$$

and put

$$F_p = \frac{y^3}{3} + y^2x^2 + \frac{yxyx}{2} + yx^4 + \frac{x^6}{6} - I_p(x).$$

Theorem 3.3. *For every cyclic formal series $h \in O_6$, the algebra*

$$A = \mathrm{Jac}(F_p + h)$$

is finite-dimensional over $\mathbb{C}$.

Suppose the Jacobi algebra were infinite-dimensional. Its associated graded algebra would be a quotient of an auxiliary algebra with a simple generic fiber of dimension 25, and this quotient would have the same generic rank. The fourth-power relation would then force $4 \mid 25$. To obtain the divisibility, we let a lift of the graded central element act on the left. Right multiplication is linear over this parameter ring even when the lift itself is noncentral.

3.3 The auxiliary graded algebra

Assign weights $1, 2$ to x, y, respectively, and define

$$Q = y^2 + yx^2 + x^2y + xyx + x^4, \qquad S = yxy + y^2x + yx^3, \qquad B = \mathbb{C}\langle x, y\rangle/(x^4, Q).$$

The free polynomial identities

$$[S, y] = y[x, Q], \qquad [S, x] = [y, Q] + x[x, Q]$$

show that S commutes with both generators modulo $Q = 0$. Thus it is central in B and has weight five.

Lemma 3.4. *The algebra B is finite over $\mathbb{C}[S]$, and*

$$\dim_{\mathbb{C}} B/(S) = 40, \qquad B/(S-1) \cong M_5(\mathbb{C}).$$

Consequently $B \otimes_{\mathbb{C}[S]} \mathbb{C}(S)$ is central simple of dimension 25.

Proof. We complete the two relations of B in weighted degree-lexicographic order. The script `nr_B5_generic_certificate.py` uses exact rational arithmetic and resolves every overlap and inclusion. With $S = 0$ it finds 40 irreducible words, and with $S = 1$ it finds 25. Both computations take place in the ordinary polynomial algebra.

For the specialization at $S = 1$, the 25-dimensional regular trace pairing has nonvanishing determinant

$$298023223876953125 = 5^{25},$$

and the center has dimension one. In characteristic zero the nondegenerate trace pairing gives semisimplicity. The center dimension then leaves a single matrix factor, necessarily $M_5(\mathbb{C})$. The bases, irreducible words and exact outputs are recorded in `nr_B5_generic_certificate.json`.

Lift homogeneous basis words of $B/(S)$. Induction on weight shows that their lifts generate B over $\mathbb{C}[S]$. To identify the generic fiber, adjoin z with $z^5 = S$ in the field extension $\mathbb{C}(S) \subset \mathbb{C}(z)$. Rescaling $x = zX$ and $y = z^2Y$ identifies $B \otimes_{\mathbb{C}[S]} \mathbb{C}(z)$ with the scalar extension of $B/(S-1)$ to $\mathbb{C}(z)$. Central simplicity and dimension 25 descend along this field extension. □

3.4 Filtered modules with a noncentral parameter

Let A be a separated complete algebra with a decreasing positive integer filtration F^nA, $F^0A = A$, finite-dimensional associated graded pieces, and $F^iAF^jA \subseteq F^{i+j}A$.

Lemma 3.5. *Suppose* $\operatorname{gr} A$ *is finite over the polynomial ring* $\mathbb{C}[\sigma]$*, where* σ *is central and homogeneous of degree* $e > 0$*. Choose a lift* $s \in F^eA$ *of* σ*. Regard* A *as a left* $D = \mathbb{C}[[t]]$*-module by* $t \cdot a = sa$*. Then* A *is a finite* D*-module and*

$$\operatorname{rank}_D A = \operatorname{rank}_{\mathbb{C}[\sigma]} \operatorname{gr} A.$$

Its D*-torsion submodule is a right ideal and is finite-dimensional over* $\mathbb{C}$*.*

Proof. Lift homogeneous $\mathbb{C}[\sigma]$-generators of $\operatorname{gr} A$ to A. Subtract left powers of s times these lifts to cancel the initial form, and repeat on the remainder. The weights increase at each step, so the coefficients converge in D. This proves finite generation. If the largest generator weight is c, the same reduction gives

$$F^{en+c}A \subseteq s^nA \subseteq F^{en}A.$$

In this reduction every term of weight at least $en + c$ uses a power of s of exponent at least n. A finite graded $\mathbb{C}[\sigma]$-module of rank r has cumulative dimension $rN/e + O(1)$ below weight N, as follows from a homogeneous basis modulo torsion. The two inclusions therefore give $\dim_{\mathbb{C}} A/s^nA = rn + O(1)$. The structure theorem over D identifies the same leading coefficient with $\operatorname{rank}_D A$.

Right multiplication commutes with left multiplication by s on A, by associativity. Hence the torsion submodule is a right ideal. As a finite torsion module over the DVR it has finite length. □

Lemma 3.6 (Filtered determinant test)**.** *In the situation of Lemma 3.5, assume the common rank is* $r > 0$*. Suppose* $z \in F^1A$*,* $m \geq 1$*,* $c \in \mathbb{C}^\times$*, and a one-variable formal series* v *satisfy*

$$q(z) = cz^m(1 + zv(z)) = s + \epsilon, \qquad \epsilon \in F^{e+1}A.$$

Then m *divides* r*.*

Proof. Write $T \subset A$ for the finite-dimensional D-torsion submodule. Then $M = A/T$ is free of rank r over D. Use the quotient filtration $F^nM = (F^nA + T)/T$. Right multiplication $R_a : m \mapsto ma$ descends because the torsion is a right ideal, and each R_a is D-linear. We now use centrality of the initial form of s in $\operatorname{gr} A$. It puts the commutator $[s, a] = sa - as$ in filtration strictly above $e + \deg(a)$, giving

$$(R_s - t\,\mathrm{id})(F^nM) \subseteq F^{n+e+1}M.$$

Thus $R_{q(z)} = t\,\mathrm{id} + E$, where the error operator E shifts the filtration by at least $e + 1$.

Adjoin u with $t = u^e$ and work over $\mathbb{C}((u))$. To read the filtration shift as a valuation, use the lattice

$$\mathcal{L} = \sum_{n \geq 0} u^{-n}\big(F^nM \otimes_D \mathbb{C}[[u]]\big) \subset M \otimes_D \mathbb{C}((u)).$$

Let $M_u = M \otimes_D \mathbb{C}[[u]]$. For $n \geq c$ the filtration bounds give $F^nM \subset t^{\lfloor (n-c)/e \rfloor}M$. Every term in the defining sum for $\mathcal{L}$ therefore lies in $u^{-c-e}M_u$, and the term at zero is M_u. Thus $\mathcal{L}$ is a full lattice.

On this lattice an operator that increases filtration by j takes values in $u^j\mathcal{L}$. In particular $E\mathcal{L} \subset u^{e+1}\mathcal{L}$, while t acts by u^e. It follows that

$$R_{q(z)} = u^e(I + uC) \quad \text{on } \mathcal{L}, \qquad v_t(\det R_{q(z)}) = r.$$

Indeed, the determinant has u-valuation er, and $v_u = e\, v_t$ on $\mathbb{C}((t))$. Since $z \in F^1A$, the operator R_z maps $\mathcal{L}$ into $u\mathcal{L}$. Thus $R_{1+zv(z)} = \mathrm{id} + R_z v(R_z)$ reduces to the identity modulo u and has unit determinant in $\mathbb{C}[[u]]^\times$. Taking determinants in $q(z) = cz^m(1 + zv(z))$ gives

$$r = m\, v_t(\det R_z).$$

The nonzero determinant $\det R_z$ belongs to $\mathbb{C}((t))$, so $v_t(\det R_z)$ is an integer. Hence m divides r. □

Remark 3.7. Left multiplication by s supplies the D-module structure, while right multiplication supplies the D-linear determinant operators.

3.5 Proof of length-five finiteness

Proof of Theorem 3.3. Write $h = g(x) + h_{\mathrm{mix}}$, where $g \in x^6\mathbb{C}[[x]]$ and every word of h_{mix} contains y. A mixed word of ordinary length at least six has $(1,2)$-weight at least seven. Absorb the pure part into $\widetilde{p} = p - g'$, whose leading term is still a_4x^4. Cyclically differentiating F_p gives

$$\partial_y F_p = Q, \qquad \partial_x F_p = S + xQ - p(x).$$

Setting the cyclic derivatives of $F_p + h$ to zero in the Jacobi algebra A, the two relations give

$$Q \in F^5A, \qquad \widetilde{p}(x) = S + (\partial_x h_{\mathrm{mix}} - x\partial_y h_{\mathrm{mix}}),$$

where the parenthesized error has weight at least six. The two relations have initial forms x^4 and Q. Hence there is a graded algebra surjection

$$B \twoheadrightarrow \mathrm{gr}\, A.$$

By Lemma 3.4, the associated graded algebra $\mathrm{gr}\, A$ is finite over the image of $\mathbb{C}[S]$.

If A were infinite-dimensional, so would be $\mathrm{gr}\, A$. As a finite module over $\mathbb{C}[S]$ it must then have nonzero generic fiber, since a finite torsion module over a polynomial ring in one variable is finite-dimensional. This fiber is a nonzero quotient of the central simple algebra $B \otimes_{\mathbb{C}[S]} \mathbb{C}(S)$ of Lemma 3.4. The quotient map has zero kernel, so the rank over $\mathbb{C}[S]$ is 25.

Apply Lemma 3.5 to the lift $s = S(x,y) \in F^5A$ of S. Lemma 3.6 now applies with $e = 5$, $z = x$, $q = \widetilde{p}$ and $m = 4$. It would force 4 to divide 25. Thus A is finite-dimensional over $\mathbb{C}$. □

3.6 Polynomial truncation and length-five realization

Lemma 3.8. *If* $\mathrm{Jac}(f)$ *has radical nilpotence exponent* L*, then replacing* f *by its ordinary truncation through any degree* $N \geq L + 1$ *leaves its closed Jacobi ideal unchanged.*

Proof. Let I and I_N denote the closed two-sided ideals generated by the cyclic derivatives of f and its degree-N truncation $f_{\le N}$, respectively. Their generators agree modulo the maximal ideal power $\mathfrak{m}^N$, and by assumption $\mathfrak{m}^L \subseteq I$. Because $N \ge L+1$, we have

$$\mathfrak{m}^L \subseteq I_N + \mathfrak{m}^{L+1}.$$

Multiply the containment by successive powers of $\mathfrak{m}$ and substitute back. This gives $\mathfrak{m}^L \subseteq I_N + \mathfrak{m}^{L+k}$ for arbitrarily large k. Closedness of I_N gives $\mathfrak{m}^L \subseteq I_N$. The differences between corresponding derivatives of f and $f_{\le N}$ belong to $\mathfrak{m}^N \subseteq \mathfrak{m}^L \subseteq I \cap I_N$, so $I = I_N$. □

First consider $G = F_p + g(x) + \ell(x)y$ with $g \in x^6\mathbb{C}[[x]]$ and $\ell \in x^5\mathbb{C}[[x]]$, and put $\widetilde{p} = p - g'$. For this two-series subfamily the free-necklace construction gives explicit transition functions on two charts. With formal coefficients these are read in the completed charts. The transition is

$$b = a^{-1}, \quad U = a^{-1}u, \quad V = a^3v + a^{-1}u^2 + u^3 + au^4 + a^2(u^5 - \widetilde{p}(u)) + a\ell(u).$$

The transition has Jacobian determinant -1. Its zero section is $\mathbb{P}^1$ with normal bundle $\mathcal{O}(1) \oplus \mathcal{O}(-3)$. The deformation algebra of this neighborhood is $\mathrm{Jac}(G)$ by [3, Corollary 6.4]. Truncation gives polynomial gluing data in this subfamily. To treat an arbitrary formal tail, we first normalize its five-jet and then use the fifth family of Wemyss's construction.

Proposition 3.9 (Length-five realization)**.** *Every potential in Theorem 3.3 has a geometric Jacobi algebra, realized by a smooth length-five cE_8 flop. Equivalently, the alternating-quartic stratum with nonzero pure fifth coefficient is realized, including all formal tails.*

Proof. The change $X = y + x^2$, $Y = x$ gives five-jet

$$\frac{X^3}{3} + \frac{(XY)^2}{2} - XY^4 - \frac{a_4}{5}Y^5.$$

At the level of five-jets, shear by $Y \mapsto Y + dX$. Restore the quartic by the quadratic X correction and remove the cyclic XX terms. This changes the coefficient B of XY^4 to $B + 5Cd$, where $C \neq 0$ is the pure Y^5 coefficient. Choose d to kill it. A quadratic Y correction removes the remaining $XYXY^2$ term. Rescaling the variables and the potential gives $X^3 + (XY)^2 + Y^5 + O_6$, with all higher terms retained in O_6.

By Theorem 3.3 the radical has a nilpotence exponent L. Truncate through degree at least $\max(5, L+1)$. Lemma 3.8 preserves the closed Jacobi ideal, and the cutoff preserves the normalized five-jet.

Apply the construction in [9, Sections 1.5–1.6, Eq. (1.3) and Theorem 1.7]. The fifth row of Theorem 1.6 lists $X^3 + (XY)^2 + Y^5 + O_6$. The realization statement following Eq. (1.3) identifies the Jacobi algebra of a listed potential with the deformation algebra of the associated two-chart neighborhood. Theorem 1.7 contracts its zero section to give a smooth length-five flop. Completing at the image of the curve and pulling back the source gives the required complete local contraction of type cE_8.

One can also check this last geometric step using the later classification. In cyclic notation put $\mathcal{N}_{2,2} = X^2Y^2 + (XY)^2/2$. Eliminating the bad quartic in $X^3 + \mathcal{N}_{2,2} + Y^5$ leaves a nonzero alternating quartic and a nonzero pure fifth coefficient. The coordinate changes of Corollary 3.2 therefore carry both five-jets to the same jet. Reading the changes in reverse on the second one gives the conversion used in [2, Lemma 8.1]. Theorem 8.3 there takes every formal tail of this second jet to an isomorphic Jacobi algebra in the fifth polynomial family of Corollary 8.14. Theorem 9.2, implication (2) ⇒ (3), realizes that family. The original proof above uses the rank obstruction, truncation and ICM gluing theorem and does not require this alternative. □

4 Growth exclusions

Set $A_f = \mathrm{Jac}(f)$ and write $J = \operatorname{rad} A_f$ for its Jacobson radical. A continuous change of coordinates with continuous inverse preserves the radical filtration and therefore leaves the dimensions $\dim_{\mathbb{C}} A_f/J^N$ invariant.

Theorem 4.1 (Zero-quartic exclusion)**.** *For every cyclic formal potential $f = x^3 + O_5$ and every $N \geq 1$,*

$$\dim_{\mathbb{C}} A_f/J^N \;\geq\; Z(N) := \sum_{k=0}^{\lfloor (N-1)/3 \rfloor} \binom{N-3k+1}{2} = \frac{N^3}{18} + O(N^2).$$

In particular $\operatorname{Jdim} A_f \geq 3$.

Theorem 4.2 (Both degenerate sextic axes)**.** *Let*

$$f = x^3 + y^4 + b(xy^2)^2 + c(xy)^3 + O_7, \qquad bc = 0,$$

with an arbitrary complete cyclic tail. Then $\operatorname{Jdim} A_f \geq 3$. *More precisely, after the weight-preserving normalization of Lemma 4.8, the following bounds hold for every positive integer N, and hence also hold for the original algebra:*

$$b = 0: \quad \dim_{\mathbb{C}} A_f/J^{3N} \geq 3\binom{N+2}{3},$$

$$c = 0: \quad \dim_{\mathbb{C}} A_f/J^{2N} \geq 2 \sum_{k=0}^{\lfloor (N-1)/2 \rfloor} \binom{N-2k+1}{2}.$$

The intersection $b = c = 0$ satisfies both bounds.

We treat the sextic axes first. Splitting a pure power produces a quiver whose truncated Jacobi algebras give the required lower bound. For the zero quartic we instead compute an associated graded algebra on an open set of fifth jets in Subsection 4.4, then specialize at each finite cutoff in Subsection 4.5.

4.1 Two complete-quiver growth bounds

Write Q_q for the quiver with q vertices and one arrow for each ordered pair of vertices, including a loop at each vertex. Let $\mathfrak{a}$ be its arrow ideal. We use the pairs $(q, m) = (3, 3)$ and $(2, 4)$, where the potential has order at least m.

For each quiver we choose a homogeneous potential for which the Hilbert series can be computed. We then vary its coefficients. To show that a finite truncation stays free over the formal parameter ring, we use the following criterion on the special fiber.

The Ginzburg dg algebra has original arrows in cohomological degree zero, opposite arrows in degree minus one, and vertex loops in degree minus two. Its zeroth cohomology is the completed Jacobi algebra. Below we also give the generators positive internal weights to take finite truncations.

Lemma 4.3 (A finite flat-complex criterion)**.** *Let $(T, \mathfrak{n}, k)$ be a Noetherian local ring and let P be a nonpositive complex of flat T-modules. Suppose $H^0(P)$ is finite and $H^{-1}(P \otimes_T k) = 0$. Then $H^0(P)$ is finite free over T.*

Proof. Put $K = \operatorname{im}(P^{-1} \to P^0)$ and $B = H^0(P) = P^0/K$. Given an element of $K \otimes_T k$ that vanishes in $P^0 \otimes_T k$, lift it to $P^{-1} \otimes_T k$. The lift is a cycle. By the hypothesis on $H^{-1}(P \otimes_T k)$ it is a boundary, so its image in $K \otimes_T k$ vanishes. Hence $K \otimes_T k \to P^0 \otimes_T k$ is injective. Flatness of P^0 now gives $\operatorname{Tor}_1^T(B, k) = 0$. Take a minimal surjection from a finite free T-module onto B. Its kernel is finite by Noetherianity and has zero reduction modulo $\mathfrak{n}$ by the Tor vanishing and minimality. Nakayama's lemma gives the claim. □

Lemma 4.4 (Exact homogeneous seeds)**.** *There are homogeneous potentials W_3 of degree three on Q_3 and W_2 of degree four on Q_2 whose Ginzburg differential graded algebras have zero negative cohomology. Their Jacobi algebras have total Hilbert series*

$$H_3(t) = \frac{3}{(1-t)^3}, \qquad H_2(t) = \frac{2}{(1-t)^2(1-t^2)}.$$

For a seed of degree m, assign internal weight one to every arrow; opposite generators have weight $m-1$ and the degree-minus-two vertex generators have weight m.

Proof. We compute first in the graded path algebras. Completion takes the product of the homogeneous components. The seed differential preserves internal weight, so these calculations also compute the completed algebras in each weight and at every finite cutoff.

For $q = 3$, let the cyclic group of order three act on the polynomial ring $\mathbb{C}[X, Y, Z]$ with characters $0, 1, 2$. The corresponding McKay quiver is the complete 3-vertex quiver Q_3. The McKay correspondence lifts the invariant cyclic potential $XYZ - XZY$ to a quiver potential whose Jacobi algebra is the skew group algebra $\mathbb{C}[X, Y, Z] \rtimes C_3$. Explicitly, taking vertex indices modulo three, let $a_i : i \to i$ denote the loops, $b_i : i \to i+1$ the forward arrows, and $c_i : i \to i-1$ the backward arrows; the cubic potential is

$$W_3 = \sum_i (a_i b_i c_{i+1} - a_i c_i b_{i-1}).$$

The cyclic derivatives give the three commutation relations with their vertex labels. The Ginzburg dg algebra is the cobar algebra of the exterior Koszul coalgebra on three generators, crossed with C_3. The Koszul resolution identifies its augmentation to the skew polynomial algebra as a quasi-isomorphism. The volume element is invariant, so the degree-minus-two generators are vertex loops. Counting PBW monomials gives the first Hilbert series.

For $q = 2$, consider the order-two group C_2 acting by the involution $X \mapsto X$ and $Y \mapsto -Y$. The McKay lift of the invariant potential $\frac{1}{2}[X, Y]^2$ is a quartic potential on the two-vertex quiver Q_2. The resulting Jacobi algebra is

$$U(\mathfrak{h}_3) \rtimes C_2, \qquad [X, Y] = Z, \quad [X, Z] = [Y, Z] = 0,$$

where $\deg X = \deg Y = 1$ and $\deg Z = 2$. The involution sends (X, Y, Z) to $(X, -Y, -Z)$. The second Hilbert series follows from PBW. To compute negative cohomology, adjoin a generator and use the stabilized potential

$$\widetilde{W} = Z[X, Y] - \tfrac{1}{2} Z^2.$$

The associated Ginzburg dg algebra is the cobar construction of the Chevalley–Eilenberg coalgebra of $\mathfrak{h}_3$, crossed with C_2. The cobar augmentation to $U(\mathfrak{h}_3)$ is an equivariant quasi-isomorphism. Indeed the PBW filtration reduces it to the exterior Koszul calculation for the

abelian three-dimensional Lie algebra. Since the differential preserves the positive internal weights, the conclusion holds separately in each weight.

To eliminate the stabilizing variables, denote the opposite generators by p, q, r, so that

$$dp = [Y, Z], \quad dq = [Z, X], \quad dr = [X, Y] - Z, \quad d\zeta = [X, p] + [Y, q] + [Z, r].$$

Set

$$Z' = Z - [X, Y], \quad p' = p + [Y, r], \quad q' = q + [r, X], \quad \zeta' = \zeta + r^2.$$

Then $dr = -Z'$, $dZ' = 0$, and

$$dp' = [Y, [X, Y]], \quad dq' = [[X, Y], X], \quad d\zeta' = [X, p'] + [Y, q'].$$

Thus the stabilized dg algebra is the quartic Ginzburg algebra with the contractible pair (r, Z') freely adjoined. Projection onto the quartic dg algebra is a quasi-isomorphism. In each internal weight, the words involving the pair form tensor complexes with a contractible factor. The coordinate change respects both the weights and the involution, proving the required vanishing. □

Proposition 4.5 (Uniform quiver bounds)**.** *Over any characteristic-zero extension of* $\mathbb{C}$*, the following hold. For every potential* $W \in \mathfrak{a}^3$ *on* Q_3*,*

$$\dim \mathrm{Jac}(W)/\mathfrak{a}^N \geq 3\binom{N+2}{3}.$$

For every potential $W \in \mathfrak{a}^4$ *on* Q_2*,*

$$\dim \mathrm{Jac}(W)/\mathfrak{a}^N \geq 2\sum_{k=0}^{\lfloor (N-1)/2 \rfloor} \binom{N-2k+1}{2}.$$

The potentials may be formal and need not be homogeneous.

Proof. Fix N and $(q, m) \in \{(3, 3), (2, 4)\}$. Only potential terms in degrees $m, \ldots, N$ affect $\mathrm{Jac}(W)/\mathfrak{a}^N$. Vary their cyclic coefficients independently about W_q over $T = \mathbb{C}[[e_1, \ldots, e_M]]$. Give the arrows, opposite arrows and vertex loops of the relative Ginzburg dg algebra over T internal weights $1, m-1, m$, respectively. The differential never lowers weight. We can therefore discard weights at least N in every cohomological degree, leaving a complex of finite free T-modules.

At the seed the differential preserves weight exactly. Its truncated complex is the direct sum of the full weight complexes below N, so Lemma 4.4 gives vanishing in degree minus one. In degree zero, the cohomology of the relative complex is the truncated Jacobi algebra $\mathrm{Jac}(W_T)/\mathfrak{a}^N$. By Lemma 4.3 it is finite free over T, of rank equal to the truncated dimension at the seed.

The differential into degree zero is a matrix of polynomials in the potential coefficients. If its generic rank is r, every $(r+1) \times (r+1)$ minor vanishes as a formal series at the seed and hence vanishes identically. At every coefficient point the rank is at most r, so the quotient dimension is at least the dimension at the seed. Translation by the seed covers the full coefficient space. Summing the coefficients of the two Hilbert series in Lemma 4.4 gives the asserted bounds. The minor identities survive scalar extension, and terms of degree greater than N do not affect this quotient. □

4.2 Growth transfer by partial morsification

We now deform the pure power so that its roots become distinct. Each root gives a vertex, and the components of the remaining variable give the arrows between vertices. To obtain a lower bound for the original algebra, we must compare the special fiber with a generic fiber that retains every vertex.

Lemma 4.6 (Pure-power splitting and nilpotent quotients)**.** *Let*

$$f = P(u) + \frac{v^{q+1}}{q+1} + g(u, v),$$

where $P \in u^m\mathbb{C}[[u]]$ and every word of g contains at least m occurrences of u, with $m \geq 3$. Suppose that, for each fixed number of u's, only finitely many words of g occur; the same conclusion holds if the substitutions below converge in the corresponding adic topology. Put $A = \mathrm{Jac}(f)$, $I = (u)$ and $J = \operatorname{rad} A$. For each N there is a finite $\mathbb{C}[[s]]$-module M_N whose special fiber is A/I^N and whose generic fiber is

$$\mathrm{Jac}(W)/\mathfrak{a}^N$$

for a potential $W \in \mathfrak{a}^m$ on Q_q over $K = \mathbb{C}((s))$. In particular,

$$\dim_{\mathbb{C}} A/J^{qN} \;\geq\; \dim_{\mathbb{C}} A/I^N \;\geq\; \dim_K \mathrm{Jac}(W)/\mathfrak{a}^N.$$

Proof. Take $F_s = f - s^q v$ over $D = \mathbb{C}[[s]]$ and let M_N be its relative Jacobi quotient modulo I^N, completed in the total (s, u, v)-adic topology. Its special fiber is A/I^N. Since $A/I = \mathbb{C}[[v]]/(v^q)$, we have $J^{qN} \subset I^N$. Put $K = \mathbb{C}((s))$ and $\lambda_i = s\zeta_q^i$ for a primitive root ζ_q.

To keep the full generic support, first work over the parameter ring. The relative relation $v^q - s^q = -\partial_v g \in I^m$ gives

$$(v^q - s^q)^{\lceil N/m \rceil} = 0 \qquad (\mathrm{mod}\ I^N).$$

Under the finite-word hypothesis the truncated derivative relations are polynomials, since only terms with at most N occurrences of u contribute. Form the polynomial quotient modulo I^N over $\mathbb{C}[[s]]$. Its reduction modulo I is the finite free algebra $\mathbb{C}[[s]][v]/(v^q - s^q)$. Each I^j/I^{j+1} is spanned by products of j copies of u separated by representatives of a basis of this algebra. The polynomial quotient is therefore finite over $\mathbb{C}[[s]]$. Modulo s its augmentation ideal is nilpotent, so the total-adic and s-adic topologies are cofinal. The quotient is already complete and equals M_N.

For convergent substitutions take the remainder of each string of v's between successive u's modulo the displayed monic polynomial in v. Its lower coefficients lie in (s), so the remainders converge s-adically and give the same finite-module argument.

Finiteness over the parameter DVR bounds the dimension of the generic fiber by that of the special fiber. To identify the generic fiber, invert s. Modulo I we obtain K^q. The powers of the factors $v - \lambda_i$ are pairwise coprime, so their Chinese remainder idempotents lift all q vertices as polynomials in v. These idempotents commute with v, although they need not commute with u.

We use the same spectral idempotents before imposing the Jacobi relations. Complete the free algebra over K at the kernel of $K\langle u, v\rangle \to K^q$. Successive lifting in the $(v^q - s^q)$-adic

topology gives the idempotents inside the closed commutative subalgebra generated by v. Relative to this copy of $K[v]/(v^q - s^q) \cong \prod_{i=1}^{q} K$, the completed free algebra has generators

$$u_{ij} = e_i u e_j \quad (1 \le i, j \le q), \qquad t_i = e_i(v - \lambda_i)e_i \quad (1 \le i \le q).$$

These are the generators of the completed path algebra on Q_q with q vertices, the q^2 arrows u_{ij}, and the additional q loops t_i. The inverse substitution sends u to $\sum_{i,j} u_{ij}$ and v to $\sum_i (\lambda_i e_i + t_i)$. The spectral projectors of this last element recover the vertex idempotents.

The expansion of the pure v-potential $\frac{v^{q+1}}{q+1} - s^q v$ at each root λ_i has nondegenerate quadratic part in the loop t_i, because

$$(v^q - s^q)'|_{v=\lambda_i} = q\lambda_i^{q-1} \neq 0.$$

Every other term in f contains at least m copies of u. Complete the squares successively in the t_i's, fixing the arrows u_{ij}, as in the noncommutative splitting lemma. The critical loop values satisfy $t_i \in (u_{ij})^m$. Substituting them therefore gives a quiver potential W of arrow order at least m, that is, $W \in \mathfrak{a}^m$.

The cyclic derivatives of the quiver potential correspond to the matrix blocks of $\partial_u F_s$ and the diagonal blocks of $\partial_v F_s$. Modulo these relations, the universal cyclic identity gives $[v, \partial_v F_s] = 0$. On any off-diagonal block with $i \neq j$, the commutator operator

$$z \longmapsto (\lambda_i + t_i)z - z(\lambda_j + t_j)$$

is invertible. Its constant term is the scalar $\lambda_i - \lambda_j = s(\zeta_q^i - \zeta_q^j) \neq 0$, and the remaining operator is topologically nilpotent. Thus the off-diagonal blocks of the derivative vanish in the quotient too. The resulting isomorphism preserves the ideal generated by u_{ij}.

Quotienting by the N-th power of the arrow ideal gives a finite-dimensional K-algebra. The substitutions identify the generic fiber of M_N with $\mathrm{Jac}(W)/\mathfrak{a}^N$. Indeed the ideal generated by u and $v^q - s^q$ is nilpotent modulo I, and $I^N = 0$, so completion at the full semisimple quotient preserves this finite fiber. Its dimension bound, together with $J^{qN} \subset I^N$, gives both inequalities. □

Remark 4.7. The finite-word hypothesis holds for the normalized potentials below. In a cyclic word avoiding xx and yyy, the number of y's is at most twice the number of x's, and the number of x's is at most the number of y's. A bound on either letter count therefore bounds the word length.

4.3 Proof for the degenerate sextic axes

Lemma 4.8 (Weighted normalization)**.** *Give x, y weights $4, 3$. Every $x^3 + y^4 + O_5$ is cyclically right-equivalent to a potential whose higher necklaces avoid cyclic xx and cyclic yyy. Below weight 24 its only possible higher necklaces are*

$$a\,xyxy^2 \quad (\text{wt } 17), \qquad b(xy^2)^2 \quad (\text{wt } 20), \qquad c(xy)^3 \quad (\text{wt } 21).$$

For $x^3 + y^4 + b(xy^2)^2 + c(xy)^3 + O_7$, the normalization leaves b, c unchanged and creates no weight-17 term.

Proof. Lemma 2.15 gives the convergent weighted normalization. For the displayed sextic jet there are no earlier bad terms to alter its coefficients. The terms of ordinary order O_7 in weights $21, 22, 23$ are all bad. Removing them changes no good necklace of those weights, so the normalized remainder still has ordinary order at least seven. □

Proof of Theorem 4.2. Normalize as in Lemma 4.8 and first suppose $b = 0$. Every mixed word in the normalized tail contains at least three x's. With only two x's a good necklace has at most four y's, hence length at most six. The only such sextic is $(xy^2)^2$, absent because $b = 0$, and the weight-seventeen necklace $xyxy^2$ is absent as well. After rescaling the coordinates we have

$$f = \frac{x^3}{3} + \frac{y^4}{4} + g, \qquad \text{where every word of } g \text{ contains at least three } x\text{'s.}$$

Apply Lemma 4.6 with retained variable $u = x$, split variable $v = y$, and $(q, m) = (3, 3)$. The bound of Proposition 4.5 yields

$$\dim_{\mathbb{C}} A/J^{3N} \geq 3\binom{N+2}{3}.$$

If $c = 0$, every mixed term of the normalized tail contains at least four y's. A good necklace with at most three y's has at most three x's and thus length at most six. The only possible sextic is $(xy)^3$, absent because $c = 0$. Now retain $u = y$ and split $v = x$. Lemma 4.6 with $(q, m) = (2, 4)$ and Proposition 4.5 give

$$\dim_{\mathbb{C}} A/J^{2N} \geq 2 \sum_{k=0}^{\lfloor (N-1)/2 \rfloor} \binom{N-2k+1}{2}.$$

Both bounds grow cubically in N, so $\operatorname{Jdim} A_f \geq 3$ on either axis. At their intersection $b = c = 0$ both bounds apply. □

4.4 PBW and syzygy lifting for the zero-quartic stratum

We next consider a vanishing reduced quartic. On a dense open set of fifth jets, we can compute the associated graded algebra even with an arbitrary higher tail. The calculation has two parts. A finite rational certificate gives the syzygies of the homogeneous relations, and exact lifts show that those relations generate the initial ideal for the perturbed potential. Specialization at each finite cutoff will extend the resulting dimension bound to the remaining fifth jets.

Put $R = \mathbb{Q}\langle x, y\rangle$ with ordinary grading, and define the homogeneous elements

$$U = yxy^2 + y^2xy, \qquad R_0 = x^2, \qquad R_1 = y^4 + xy^2x + yxyx + xyxy.$$

Let $M = RpR \oplus RqR$ be the free relation bimodule with generator weights $2, 4$. The map $d_0 : M \to R$ sends p to R_0 and q to R_1.

Lemma 4.9 (Finite rational syzygy certificate)**.** *The kernel of d_0 is generated as an R-bimodule by*

$$\begin{aligned} s_1 &= xp - px, \\ \sigma &= x(yq - qy) + (yq - qy)x - Up + pU, \end{aligned}$$

and all Peiffer identities

$$R_i a\theta_j - \theta_i a R_j, \qquad i, j \in \{0, 1\}, \quad a \in R, \quad \theta_0 = p,\ \theta_1 = q.$$

Moreover

$$H_{R/(R_0,R_1)}(t) = \frac{1}{(1-t)^2(1-t^3)}.$$

Proof. We equip the free algebra with the degree-lexicographic term order with $x < y$. A homogeneous Gröbner basis for the two-sided ideal is

$$\begin{aligned}
&x^2,\\
&y^4 + yxyx + xy^2x + xyxy,\\
&y^3xy^2x + y^3xyxy - yxyxy^3 - xy^2xy^3,\\
&y^3xyx + y^2xy^2x + y^2xyxy - yxyxy^2 - xy^2xy^2 - xyxy^3,\\
&y^2xyx + yxy^2x - xy^2xy - xyxy^2.
\end{aligned}$$

The fourth basis element is redundant and is retained in the certificate. The Python verifier checks that all five elements lie in the ideal of the two original relations and resolves all seventeen overlaps and inclusions. Substituting the recorded expressions back into the free relation bimodule gives eleven nonzero critical syzygies. Their weights and multiplicities are

weight	3	6	7	8	9	10
number	1	1	1	4	3	1.

The JSON certificate expresses these eleven critical syzygies using thirty-one explicit rational multiples of s_1, σ, and Peiffer identities. The verifier checks these identities in the free relation bimodule over the polynomial free algebra, using exact rational coefficients.

To see why the critical syzygies suffice in every weight, take a polynomial identity among the relations and compare two occurrences at its largest leading word. Intersecting occurrences give an overlap or inclusion identity. Disjoint occurrences give a Peiffer identity. Subtracting these comparisons lowers the leading word, so the reduction terminates. Replace each added Gröbner generator by its recorded expression in the two original relations. Its defining identity then vanishes, and expanding its Peiffer identities gives sums of Peiffer identities for the original generators. This proves the kernel assertion.

From the leading monomials of the Gröbner basis, the minimal forbidden words are

$$xx, \qquad yyyy, \qquad yyxyx, \qquad yyyxyyx.$$

Their prefix automaton has ten states. The certificate supplies exact polynomial numerators $n_i(t)$ with common denominator $D(t) = (1+t)(1-t)^2(1-t^3)$ satisfying the equations

$$n_i(t) = D(t) + t \sum_{i \to j} n_j(t).$$

The numerator at the empty state is $1 + t$. Since the automaton equations have a unique formal solution, these verified polynomial identities prove the asserted rational Hilbert series in every degree. □

Tensor each homogeneous component of $M \xrightarrow{d_0} R$ with $\mathbb{C}$ over $\mathbb{Q}$. Flatness preserves the kernel in each degree, so the displayed syzygies and their homogeneous bimodule multiples also generate the kernel over the complex numbers. We write $\widehat{R} = \prod_{n \geq 0} R_n$ for degree completion, with ground field $\mathbb{C}$ after this scalar extension. In the next lemma we reduce modulo one finite power of the augmentation ideal at a time.

Lemma 4.10 (Liftable syzygies and a closed initial ideal)**.** *Let r_i be filtered formal relations with homogeneous initial relations R_i. If every homogeneous syzygy of the R_i is a sum of homogeneous syzygies admitting exact syzygy lifts for the r_i with the same initial terms, then*

$$\operatorname{gr}(\widehat{R}/\overline{(r_i)}) \cong R/(R_i).$$

Proof. Let a nonzero element of the closed relation ideal have order n. Modulo order $n+1$ choose a representation by relation multiples. If its lowest-weight part cancels, that part is a homogeneous syzygy of the R_i. Subtract an exact lift of the syzygy. The represented element stays fixed while the least weight in its representation increases. After finitely many steps the weight-n initial form is a combination of the R_i. Hence the initial ideal is contained in (R_i). Conversely, it contains each R_i as the initial form of r_i. This argument applies to the closed ideal because it only requires a representation through the order of the element in question. □

Proposition 4.11 (Generic fifth-jet normal form)**.** *For every cyclic $h \in O_6$, let*

$$f = \frac{x^3}{3} + xyxy^2 + \frac{y^5}{5} + h.$$

Then

$$\operatorname{gr}_J \operatorname{Jac}(f) \cong \frac{\mathbb{C}\langle x, y\rangle}{(x^2,\ y^4 + xy^2x + yxyx + xyxy)}.$$

In particular, $\operatorname{gr}_J \operatorname{Jac}(f)$ *has Hilbert series* $1/((1-t)^2(1-t^3))$.

Proof. The two cyclic derivatives are

$$r_x = x^2 + U + \partial_x h, \qquad r_y = R_1 + \partial_y h.$$

Assign weights $2, 4$ to the formal relation symbols p, q. The universal cyclic identity gives the exact syzygy

$$S_1 = [x, p] + [y, q], \qquad dS_1 = [x, r_x] + [y, r_y] = 0.$$

The initial term is s_1 of weight three. From this lift form a second syzygy

$$S_2 = xS_1 + S_1x - r_xp + pr_x.$$

Indeed, direct evaluation shows $dS_2 = 0$. Cancelling its weight-four terms gives

$$S_2 = \sigma - (\partial_x h)p + p(\partial_x h).$$

Since $\partial_x h$ has order at least five, the last two terms have weight at least seven. Thus S_2 lifts the weight-six syzygy σ. Replacing R_i by r_i lifts every Peiffer identity. After extension to the complex numbers, Lemma 4.9 gives all homogeneous syzygies over $\mathbb{C}$. Lemma 4.10 then identifies the associated graded algebra.

□

4.5 Semicontinuity for the zero-quartic stratum

Proof of Theorem 4.1. Scale the potential so its cubic part is $x^3/3$. In degree five an order-three correction of x removes every necklace containing xx. The remaining fifth jet is

$$\frac{x^3}{3} + Axyxy^2 + Bxy^4 + Cy^5.$$

When $C \neq 0$, apply the shear $y \mapsto y + tx$ and remove the cyclic xx terms it creates. The surviving coefficients become

$$A' = A + 2Bt + 5Ct^2, \qquad B' = B + 5Ct, \qquad C' = C.$$

The shear preserves $B^2 - 5AC$. On the open set

$$C \neq 0, \qquad B^2 - 5AC \neq 0,$$

take $t = -B/(5C)$. This kills the xy^4 term, since $B' = 0$, and leaves $A' = -(B^2 - 5AC)/(5C) \neq 0$. Rescale x, y and the potential to obtain the fifth jet of Proposition 4.11. No quartic is introduced, and the remaining O_6 tail is arbitrary. These changes preserve the radical filtration, so Proposition 4.11 gives the same truncated dimensions throughout this open set.

Fix N. Vary the coefficients through degree $\max(N, 5)$, so the fifth-jet conditions are defined even for small N. Modulo $\mathfrak{m}^N$ the two-sided multiples of the derivatives give a finite relation matrix polynomial in the coefficients. Its rank on the dense open set is a constant r, since Proposition 4.11 gives the same cumulative Hilbert function for every tail there. Each minor of size $r + 1$ vanishes on that set and hence identically. Every specialization therefore has quotient dimension at least the generic value. The derivatives of terms beyond degree N lie in $\mathfrak{m}^N$, so this proves the bound for the complete potential.

Finally, the degree-n coefficients of the Hilbert series are given by

$$[t^n]\frac{1}{(1-t)^2(1-t^3)} = \#\{(a, b, c) \in \mathbb{Z}^3_{\geq 0} : a + b + 3c = n\}.$$

Summing over $n < N$ gives $Z(N)$. The leading quadratic terms of the binomial coefficients sum to $N^3/18 + O(N^2)$. □

5 Proof of the realization theorem

5.1 The growth-one model

The length-three calculation leaves a single candidate for the growth-one algebra,

$$\mathrm{Jac}(x^3 + xy^3).$$

To realize it, choose the following arc in the marked E_6 family. Its equations are

$$\beta^3 = w, \qquad \gamma^3 = -w, \qquad \delta^2 = -w/3, \qquad \beta + \gamma + \delta = 0.$$

The constant coefficients of all three polynomials have nonzero first derivatives, so the source is regular. The two cubics have opposite root sets, giving the persistent coefficient-two roots

and hence a divisor-to-curve contraction. Its three-direction potential is $-\delta^3+\beta^4/4-\gamma^4/4$. Substituting $x=\beta+\gamma$ and $y=(\beta-\gamma)/2$ gives the cyclic class

$$x^3+xy^3+\tfrac{1}{4}x^3y.$$

It remains to check that eliminating the last term creates no pure y-series. Let $\iota(x)=-x$ and $\iota(y)=y$. The seed and the error are odd under ι. At each weight apply $(1-\iota)/2$ to the tangent decomposition in Lemma 2.12. Since r_x is even and S is odd, this gives an odd x-correction and an even y-correction and kills the pure y-term. The substitution commutes with ι, so the next error is again odd. The weight-adic limit gives x^3+xy^3 and identifies the completed Jacobi algebra of the arc.

The nonzero pure-series perturbations are finite-dimensional by Theorem 2.13. This accounts for the growth-one member. For another divisor-to-curve model of the same algebra, see [2, Theorem 1.7].

5.2 Completing the quartic division

Proof of Theorem 1.3. By Proposition 2.1 it suffices to realize the potentials x^3+O_4 satisfying the growth bound. Apply the quartic division of Proposition 2.3.

If $\Delta\neq 0$, Theorem 2.13 constructs the length-three arc and distinguishes the small and divisorial cases.

For an alternating quartic, Theorem 3.1 forces a nonzero pure fifth coefficient. Normalize the five-jet by Corollary 3.2. The change $X=y+x^2$, $Y=x$ preserves ordinary order in both directions, so in the other coordinates the potential has the form F_p+h with $p(u)=u^4$ and $h\in O_6$. Theorem 3.3 proves finiteness, and Proposition 3.9 realizes the algebra in length five.

For the pure y^4 quartic, Lemma 4.8 isolates the coefficients of weights $17, 20, 21$. If the weight-17 coefficient is nonzero, Theorem 2.16 applies. If it vanishes, Theorem 4.2 excludes either sextic coefficient being zero, leaving exactly the nonzero pair of Theorem 2.17. The remaining quartic orbit, the zero quartic, is excluded by Theorem 4.1.

Generic surface types. Varying a hyperplane in the fixed threefold deforms its distinguished partial-resolution surface and preserves the scheme fiber. The generic Du Val diagram is a subgraph of the distinguished one with the same retained-cycle multiplicity. For the length-three E_6 family no proper subgraph has multiplicity three. Thus the generic type is cE_6, also for the divisorial member. For the small primitive contractions of lengths four, five and six, the invariant-theoretic criteria of [6, Section 8] give cE_7, cE_8, cE_8, respectively.

Under Jdim ≤ 1 the surviving Type E branches are therefore

normalized branch	construction	generic surface type
$\Delta\neq 0$	marked length 3	cE_6
$(xy)^2$ quartic, nonzero pure fifth	marked length 5	cE_8
y^4 quartic, $a_{17}\neq 0$	marked length 4	cE_7
y^4 quartic, $a_{17}=0,\ b_{20}c_{21}\neq 0$	marked length 6	cE_8

The growth bounds exclude every other branch.

In lengths three, four and six, Lemma 2.5 identifies the normalized basic contraction quotient. For length five, Lemma 3.8 identifies the closed Jacobi ideal of the polynomial used

in Proposition 3.9. All the changes of coordinates and scalar multiples transport the completed Jacobi algebra isomorphically back to the original potential.

Together with the Type A and D cases of Proposition 2.1, these constructions satisfy Definition 1.1 for every algebra in the theorem. The finite-dimensional algebras are realized by flops and the growth-one algebras by divisor-to-curve contractions. □

6 Computational certificates

The computations use exact arithmetic. Starting in the manuscript directory, run

```
cd tools/numerical_realisation_certificates
python3 verify_cyclic_reductions.py
python3 verify_zero_quartic.py
python3 nr_B5_generic_certificate.py --output ../../build/nr_B5_rerun.json
```

The first two programs use the Python standard library. The third uses the versions of SymPy and mpmath listed in `requirements-certificates.txt` in the manuscript directory.

For Lemma 2.12, the first script reduces each of the ten overlaps in both orders. For Theorem 2.17 it forms the ordinary and cyclic relation matrices of B_6, computes their ranks over the rationals, and checks the seven proposed cocenter generators.

The second script reads the adjacent `zero_quartic_certificate.json`. It checks the five original-ideal identities, the seventeen ambiguities and the thirty-one summands of the eleven critical syzygies in Lemma 4.9. It also verifies the polynomial identities giving the Hilbert series of the prefix automaton. Lemma 4.10 passes from the homogeneous relations to formal relations.

The third script recomputes both specializations in Lemma 3.4. The supplied `nr_B5_generic_certificate.json` records the basis words, the dimensions 40 and 25, and the regular trace determinant and center dimension at $S = 1$. These are the finite inputs to the filtered argument of Theorem 3.3.

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For this reason, I do not wish the existence or priority date of this manuscript to discourage a genuinely independent human derivation of these results. If a subsequent work obtains the same results by an independently developed, entirely human mathematical argument, I make no claim to special priority or intellectual credit for the corresponding proof ideas beyond the ordinary bibliographic acknowledgement of prior or concurrent results.

If, however, a subsequent argument makes substantive use of the results, constructions, or proof strategies contained in this manuscript, ordinary scholarly citation remains appropriate regardless of the tools used. In particular, when closely related arguments arise in AI-assisted work, I would appreciate explicit attribution, since the provenance of ideas generated through overlapping language-model systems can otherwise be difficult to determine.

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